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Dynamic flexibility management for pumped hydro storage: Enhancing frequency stability and economic efficiency in cascade hydro-wind-solar-storage integrated delivery systems

Xingjin Zhang [a,b], Jizhou Wang [a,b], Ziwen Zhao [a,b], Fangfang Li [c], Pin Jern Ker [d], Ali Najah Ahmed [e], Beibei Xu [a,b], Diyi Chen [a,b,*]

[a] Key Laboratory of Agricultural Soil and Water Engineering in Arid and Semiarid Areas, Ministry of Education, Northwest A & F University, Shaanxi, Yangling 712100, PR China
[b] Institute of Water Resources and Hydropower Research, Northwest A&F University, Shaanxi, Yangling 712100, PR China
[c] College of Water Resources & Civil Engineering, China Agricultural University, Beijing 100083, PR China
[d] School of Engineering and Technology, Sunway University, Bandar Sunway, Petaling Jaya 47500, Malaysia
[e] Department of Engineering, School of Engineering and Technology, Sunway University, No:5, Jalan Universiti, Bandar Sunway, 47500, Selangor Darul Ehsan, Malaysia

A R T I C L E I N F O

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A B S T R A C T

The integration of substantial variable renewable energy (VRE) into the cascade hydro-wind-solar-storage integrated delivery system (CHIDS) poses significant challenges in maintaining frequency stability and optimizing economic performance. This study proposes a balance-oriented strategy for the flexible operation of pumped hydro storage (PHS) to address these issues, enhancing both frequency stability and economic efficiency. A quantitative method is developed to evaluate flexibility supply capacity, incorporating both fixed-speed (FS-PHS) and variable-speed (VS-PHS) units. A multi-level flexibility balance constraint model is introduced to facilitate steady-state power regulation, transient frequency response, and quasi-steady-state frequency recovery. Additionally, a multi-timescale scheduling strategy is employed to optimize power delivery by balancing frequency stability and economic performance through flexibility constraints. Results indicate that the proposed strategy reduces the probability of insufficient flexibility by 15 % and 4 % for the CHIDS involving FS-PHS and VS-PHS, respectively, compared to conventional frequency security-based scheduling. Furthermore, the strategy ensures frequency stability, with recovery probabilities stabilizing at 98 %. At a VRE penetration of 62 %, the system achieves optimal performance, with delivery reliability approaching 99 % and curtailment reduced to 3 %. Compared to strategies that do not account for frequency security constraints, the internal rate of return increases by 0.08 % and 0.24 % for the FS-PHS and VS-PHS cases, respectively. These findings underscore the critical need to balance flexibility supply and demand to improve system reliability and economic efficiency in future high-VRE systems.

1. Introduction

1.1. Research motivation

As global energy systems undergo rapid transformation, newly installed renewable energy capacity exceeded 700 GW in 2024, with wind and PV power comprising 95 % of the overall capacity expansion [1]. However, the integration of a high proportion of variable renewable energy (VRE) into the grid has induced dynamic challenges in the power delivery system across multiple timescales. On a second timescale, stochastic fluctuations inherent to wind and PV power reduce system inertia and heighten the risk of frequency deviations [2,3]. Conversely, on an hourly timescale, spatial–temporal mismatches between generation and load decrease the utilization efficiency of interregional transmission corridors, thereby aggravating curtailment rates and supply deficits [4]. Pumped storage hydropower (PSH), constituting over 90 % of installed storage capacity, has emerged as a critical technology [5]. Through its bidirectional power modulation and rapid frequency response capabilities, PSH has become the central hub for balancing safety and economic performance within the cascaded hydro-wind-

* Corresponding author at: Institute of Water Resources and Hydropower Research, Northwest A&F University, Shaanxi, Yangling 712100, PR China.
E-mail address: diyichen@nwsuaf.edu.cn (D. Chen).

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solar-storage integrated delivery system (CHIDS) [6]. By 2024, China's PSH capacity reached 58 GW, underscoring its essential role in mitigating fluctuations, ensuring frequency stability, and enhancing transmission efficiency within CHIDS.

Despite the maturation of PSH technology, high-VRE penetration systems present three primary challenges to operational flexibility. First, traditional fixed-speed (FS-PHS) units, limited by constant rotational speeds, are insufficient for responding to sub-second frequency fluctuations. Although variable-speed (VS-PHS) units provide rapid adjustment capabilities, their frequent start–stop cycles and operational mode transitions significantly increase mechanical stress and maintenance costs [7]. Empirical studies indicate that for every 5 % increase in VRE penetration, the regulation capacity of PSH must be augmented by 2–5 % to sustain system stability [8]. Second, conventional single-machine equivalent models overlook dynamic disparities—such as differences in frequency regulation deadband and inertial response—between FS-PHS and VS-PHS units, leading to an overestimation of system delivery reliability and VRE absorption potential [9,10]. Finally, the relationship between VRE penetration and system dynamic performance is highly nonlinear. Existing studies, predominantly based on static assumptions, lack a dynamic analysis of critical penetration thresholds. Exceeding these thresholds may precipitate a precipitous deterioration in frequency security due to exhausted PSH regulation margins, while transmission corridor utilization becomes constrained by risks associated with power fluctuations surpassing established limits [11,12]. These challenges underscore the urgent need to develop a multi-timescale coupled optimization framework that integrates precise dynamic characterization of generation units to enable closed-loop feedback for dispatch optimization and threshold decision-making.

1.2. Literature review

Existing research primarily constructs flexibility quantification models for PSH units from two perspectives: steady-state flexibility provision and transient dynamic support. At the steady-state level, scholars have concentrated on characterizing power regulation over time scales from minutes to hours. Commonly employed indicators include ramp rate, power band, and cycling capability to evaluate flexibility supply efficacy. For instance, Chen et al. [13] proposed an optimization model for regional flexibility matching, wherein the first derivative of the net load defines the baseline flexibility demand, and a mixed-integer programming approach optimally matches multiple PSH sites with regional demand. To address the quantification challenges of heterogeneous unit coordination, Zhao et al. [9] developed a multidimensional model that integrates minimum stable generation/pumping power constraints, turbine-pump efficiency disparities, and unit configuration optimization, thereby elucidating the influence of subsidy policies on economic and environmental performance. Similarly, Wu et al. [14] decoupled operational from strategic flexibility, constructing models to evaluate power regulation accuracy based on vibration zone avoidance and economic potential through unit combination optimization. In the realm of transient frequency support modeling, the focus shifts to sub-second dynamic characteristics. Key metrics include inertia time constant, rate of change of frequency, and frequency regulation delivery time. Lan et al. [15] established a hydraulic-mechanical coupled optimization framework and revealed nonlinear relationships among guide vane closure patterns, runner geometry, and transient stability through sensitivity analysis. Zhao et al. [16] introduced a three-stage control optimization framework that dynamically balances model precision, convergence efficiency, and conflicting objectives, with field tests demonstrating a 9.6 % improvement in tailrace vacuum. Yang et al. [17] further quantified the dynamic coupling between frequency regulation quality and economic indicators, such as efficiency losses and mechanical fatigue. Despite these advances, two critical deficiencies persist: (1) prevailing linear aggregation models overlook the nonlinear coupling between FS-PHS and VS-PHS units in terms of regulation range, inertial response, and frequency regulation capacity; and (2) existing models fail to integrate transient safety constraints with steady-state economic objectives, rendering the trade-offs between unit regulation losses and system regulation margins inadequately quantified.

Furthermore, operational optimization research has increasingly emphasized multi-timescale planning. Daneshvar et al. [18] developed a two-stage stochastic programming model that utilizes nonlinear optimization techniques to maximize day-ahead profits for PHS. Wang et al. [19] devised a scheduling strategy coupling hydraulic characteristics with reliability, thereby sustaining system efficiency above 90 % and achieving a VRE accommodation rate of 95.47 %. Han et al. [20] designed a coordinated controller for load following and ancillary services, which, although effective in mitigating seasonal wind curtailment, failed to establish a mapping between transient dynamics and dispatch decisions. Notably, Alizadeh-Mousavi et al. [21] formulated a stochastic security-constrained unit commitment (SCUC) model that minimizes scheduling costs by differentiating the energy-reserve supply characteristics of FS-PHS and VS-PHS units. Lin et al. [22] proposed a joint dispatch method for PSH and renewable energy units, incorporating constraints on frequency deviation and voltage stiffness, and successfully maintained frequency deviations within 0.5 Hz in an enhanced IEEE 39-bus system. Han et al. [23] developed an optimization framework spanning from 15-min to second-level timescales, quantifying hydropower flexibility supply costs in terms of hydraulic energy losses. Zheng et al. [24] introduced an integrated control optimization approach that simultaneously considers economic costs and closed-loop dynamic performance; although it reduced electrical deviations by 28.11 %, its treatment of nonconvex constraints resulted in elevated computational complexity. A common limitation among these studies is the absence of a quantitative linkage between the transient frequency safety domain and quasi-steady-state recovery capability, thereby exposing the system to dual pressures of frequency excursion risks and rising regulation costs.

In the realm of VRE penetration threshold decision-making, prevailing methodologies adhere to either a techno-economic or dynamic frequency safety approach. The techno-economic path, centered on the levelized cost of energy (LCOE), was demonstrated by Hassan et al. [25] via comparisons under 40 % and 70 % VRE penetration, revealing marginally increasing LCOE with higher penetration; however, its static sensitivity analysis fails to capture the time-varying regulation capabilities of units. In contrast, the dynamic safety approach emphasizes frequency stability constraints. Luo et al. [26] developed a multi-source frequency response model indicating that, with renewables constituting 80 % of total generation, the system can tolerate disturbances amounting to 10–15 % of the load—albeit without considering PSH regulation capacity constraints. Song et al. [27] further proposed a multi-objective capacity configuration model, employing the NSGA-II algorithm to derive a Pareto frontier that balances minimized LCOE with maximized effective load carrying capability, yet it does not quantify the coupling effects between system frequency dynamics and unit regulation capabilities. Although Perera et al. [28] utilized reinforcement learning for intelligent configuration of distributed energy systems, the opaque nature of the approach undermines its interpretability.

In summary, several critical limitations persist that require resolution: (1) existing unit group flexibility models frequently overlook the nonlinear disparities in regulation range, and frequency regulation capacity between FS-PHS and VS-PHS units, with linear aggregation methods inducing systematic bias; (2) traditional dispatch models segregate day-ahead economic optimization from intra-day frequency control, lacking a comprehensive "steady-state–transient–quasi-steady-state" constraint framework to address the inverse impact of second-level disturbances on hourly dispatch plans; and (3) prevailing transmission capacity assessments, based on static sensitivity analyses, fail to establish a quantitative model for the dynamic coupling between frequency processes and time-varying unit regulation capabilities, resulting in a dual dilemma of overly conservative and insufficiently robust capacity decisions.

1.3. Contributions and paper organization

To address the aforementioned research gaps, this study introduces an innovative framework that integrates differentiated unit modeling, multi-timescale nested optimization, and dynamic coupling assessment, thereby resolving the conflict between PHS flexibility and system economic performance under high renewable penetration scenarios. The primary contributions are as follows:

(1) A differentiated operational model for FS-PHS and VS-PHS units is developed, establishing a novel three-dimensional flexibility balancing constraint framework that concurrently encompasses steady-state power regulation, transient frequency support, and quasi-steady-state recovery, thereby revealing inherent nonlinearities in heterogeneous flexibility provision.
(2) A scenario-reduction-based optimization scheduling model is proposed, wherein transient frequency safety constraints are mapped onto day-ahead and intra-day scheduling variables via mixed-integer piecewise linearization. This model synergistically optimizes steady-state economic objectives and transient safety requirements. Compared to traditional SCUC frameworks, its nested structure reduces problem dimensionality, while dynamic reserve capacity partitioning minimizes regulation losses, enhances computational efficiency, and balances safety margins with operational economy.
(3) A comprehensive framework is established to assess flexibility from the perspectives of frequency safety, transmission reliability, and economic performance. This study further proposes a VRE carrying capacity decision framework. By quantitatively elucidating the nonlinear impact of renewable penetration on system performance, the framework facilitates the dynamic alignment between renewable installed capacity and pumped storage regulation capabilities.

The remainder of this paper is organized as follows: Section 2 introduces the quantitative evaluation method for assessing the flexibility supply of PHS units and presents the multi-timescale optimization scheduling model. Section 3 describes the optimization framework for determining VRE integration capacity. Section 4 details the case study of a clean energy hub in Qinghai Province, and Section 5 presents and discusses the results. Finally, conclusions are shown in Section 6.

2. Methodology

2.1. Description of the cascade hydro-wind-solar-storage integrated delivery system

The structure of the CHIDS is illustrated in Fig. 1. It comprises four subsystems—hydropower, wind power, photovoltaic generation, and PHS. Detailed output models for each subsystem are provided in Appendix A. Electrical output from these subsystems is consolidated at the aggregation station and transmitted to the distant load center of the receiving grid via an ultra-high-voltage direct-current (UHVDC) transmission link forming a prototypical point-to-grid interconnection topology. Although the system is electrically coupled to the main grid through UHVDC, its frequency security must be autonomously guaranteed by local flexibility resources for the following reasons: 1) DC transmission cannot convey inertial response or dynamic frequency support. The sending-end CHIDS must independently counteract frequency-instability risks arising from fluctuations in renewable output or DC link outages. 2) The pronounced variability of wind and PV generation significantly diminishes the system's equivalent inertia, thereby mandating the provision of synthetic inertia, peak-shaving, and frequency regulation services by cascade hydropower and pumped-storage assets. 3) In the event of a UHVDC link outage, the source-side system must instantaneously transition to self-sufficient frequency control modes to avert frequency collapse, akin to a quasi-islanded operational state.

2.2. Operational flexibility of the pumped hydro storage

In power systems, operational flexibility is defined as the integrated capability to offset power imbalances and mitigate frequency deviations induced by the variability of wind and solar energy, achieved through the coordinated utilization of controllable resources' power regulation capacities and rapid response characteristics [29]. This concept encompasses two principal dimensions: (i) power regulation flexibility, which characterizes the system's ability during the steady-state phase (ranging from minutes to hours) to adapt to fluctuations in renewable output via parameters such as unit ramp rates and operational output ranges; and (ii) frequency response flexibility, which denotes the dynamic capacity of the system during the transient phase (ranging from seconds to minutes) to suppress frequency disturbances through mechanisms including inertia support and frequency regulation reserves.

To elucidate the mechanisms underlying the contribution of PHS units to system flexibility, this section initiates an analysis from the perspective of evolving functional roles within modern power systems. It

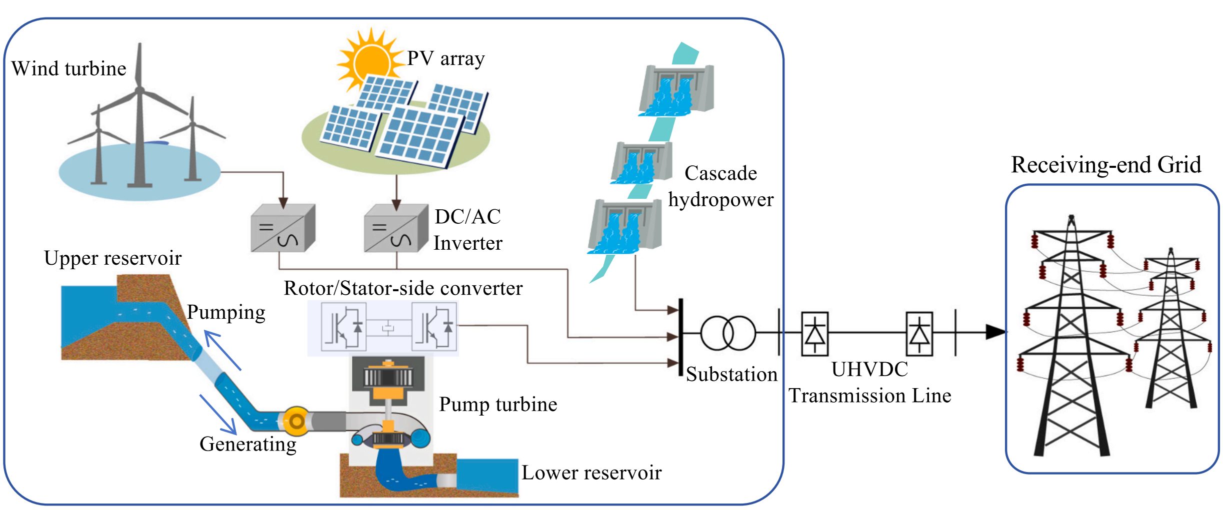


Fig. 1. Overall structure of the cascade hydro-wind-solar-storage integrated delivery system.

further contrasts the disparities in power modulation and frequency response capabilities between fixed-speed and variable-speed PHS units, ultimately quantifying their flexibility provision efficacy through mathematical modeling.

2.2.1. Power regulation of pumped hydro storage units

Conventional primarily serve the function of peak shaving and valley filling (Fig. 2a), with operational strategies tailored to the load curve's peak-valley characteristics. However, within the CHIDS, the pronounced stochasticity of VRE generation, coupled with the spatiotemporal imbalances inherent in interregional transmission corridors, necessitates an evolution in the operational role of PHS units. In addition to peak shaving and valley filling, PHS units must also implement mid-load demand to supply remote load centers, effectively balancing the variability associated with high proportions of VRE generation (Fig. 2b). Specifically, when transmission capacity is sufficient and VRE generation is low, the PHS units generate electricity, swiftly addressing the need for stable power delivery; conversely, during peak VRE generation and interregional transmission corridors near saturation, these units switch to pumping mode to absorb surplus power and avert transmission overload.

Additionally, PHS units are classified into fixed-speed and variable-speed types based on their electromagnetic coupling mechanisms, with each exhibiting fundamentally distinct power regulation capabilities, as shown in Fig. 3. FS-PHS units are constrained by the inherent rigidity of synchronous machine connections, typically confining their power modulation range in generation mode to between 40 % to 100 % of rated capacity [21]. In contrast, VS-PHS units employing doubly-fed induction generators or full-power converters leverage electromechanical decoupling control to extend their modulation range to between 20 % and 100 % of rated power, while obviating the need for mechanical braking during power reversal [21].

FS-PHS units operating in generation mode exhibit an upward regulation capacity that is limited by the real-time power output and the maximum ramping rate, as illustrated in Eq. (1), while, the downward regulation reserve is constrained by the minimum technical output, as specified in Eq. (2).

$$R_{up,t}^{PS.G} = \sum_{g \in N_{PS}} \min\left(0, \Delta P_{ramp.g}^{PS} \cdot \Delta t,\ \ u_{g,t}^{PS.G} \cdot P_{g,\max}^{PS.G} - P_{g,t}^{PS.G}\right) \tag{1}$$

$$R_{dn,t}^{PS.G} = \sum_{g \in N_{PS}} \min\left(0, \Delta P_{ramp.g}^{PS} \cdot \Delta t,\ \ u_{g,t}^{PS.G} \cdot P_{g,t}^{PS.G} - P_{g,\min}^{PS.G}\right) \tag{2}$$

where N_{PS} denotes the number of units within the PHS station, $P_{g,t}^{PS.G}$ and $u_{g,t}^{PS.G}$ are the power generation and state variable of unit g at time t. $\Delta P_{ramp,g}^{PS}$represents the ramp-up rate, Δt the scheduling time step, $R_{up,t}^{PS.G}$ and $R_{dn,t}^{PS.G}$ are the upward and downward regulation reserve of unit g in the generation mode at time t, respectively. $P_{g,min}^{PS.G}$ and $P_{g,max}^{PS.G}$ are the minimum and maximum generation power outputs, respectively. Notably, under pumping mode, FS-PHS units are restricted to operating solely at their rated power, thereby exhibiting an absence of modulation flexibility. Consequently, the upward reserve capacity under these conditions is realized exclusively through emergency shutdown procedures.

In VS-PHS units, the computation of upward and downward regulation reserve capacities in generation mode is analogous to that for FS-PHS units, as outlined in Eqs. (1) and (2). Under pumping mode, the regulation range of VS-PHS units typically spans from 50 % to 100 % of rated power. Specifically, the upward regulation capacity is achieved by reducing pumping power (Eq. (3)), whereas the downward regulation capacity is realized by increasing pumping power (Eq. (4)).

$$R_{up,t}^{PS.P} = \sum_{g \in N_{PS}} \min\left(0, \Delta P_{ramp.g}^{PS} \cdot \Delta t,\ \ u_{g,t}^{PS.P} \cdot P_{g,\max}^{PS.P} - P_{g,t}^{PS.P}\right) \tag{3}$$

$$R_{dn,t}^{PS.P} = \sum_{g \in N_{PS}} \min\left(0, \Delta P_{ramp.g}^{PS} \cdot \Delta t,\ \ P_{g,t}^{PS.P} - u_{g,t}^{PS.P} \cdot P_{g,\min}^{PS.P}\right) \tag{4}$$

where $P_{g,t}^{PS.P}$ and $u_{g,t}^{PS.P}$ are the pumping power and state variable of unit g at time t. $R_{up,t}^{PS.P}$ and $R_{dn,t}^{PS.P}$ denote the upward and downward reserve capacities of unit g in the pumping mode at time t, respectively. $P_{g,min}^{PS.P}$ and $P_{g,max}^{PS.P}$ are the minimum and maximum pumping power, respectively.

Based on the aforementioned model analysis, the flexibility provision efficacy of PHS in the CHIDS mode is predominantly manifested in its bidirectional, self-adaptive modulation capability in response to grid power fluctuations. Upward regulation flexibility is principally engaged during periods of diminished VRE output or peak load conditions, wherein the units either enhance generation power or curtail pumping power to supplement the electrical supply. Conversely, downward regulation flexibility serves to attenuate surplus VRE, achieved through an increase in pumping load or a reduction in generation power to absorb the excess energy. Moreover, to ensure the reliability of peak regulation reserve, the PHS facility must maintain sufficient water storage and inventory capacity to meet modulation requirements over varying temporal scales. Therefore, the aggregate upward and downward flexibility provision can be respectively expressed as:

$$R_{up,t}^{PS} = \min\left(R_{up,t}^{PS.G},\ \ R_{dn,t}^{PS.P},\ \frac{V_t^{up} - V_{\min}^{up}}{\xi_G^{PS} \Delta t}\right) \tag{5}$$

$$R_{dn,t}^{PS} = \max\left(R_{dn,t}^{PS.G},\ \ R_{up,t}^{PS.P},\ \frac{V_{\max}^{up} - V_t^{up}}{\xi_P^{PS} \Delta t}\right) \tag{6}$$

where $R_{up,t}^{PS}$ and $R_{dn,t}^{PS}$ are the upward and downward reserves of the PHS station at time t, respectively. V_t^{up}, V_{min}^{up}, and V_{max}^{up} represent the upstream

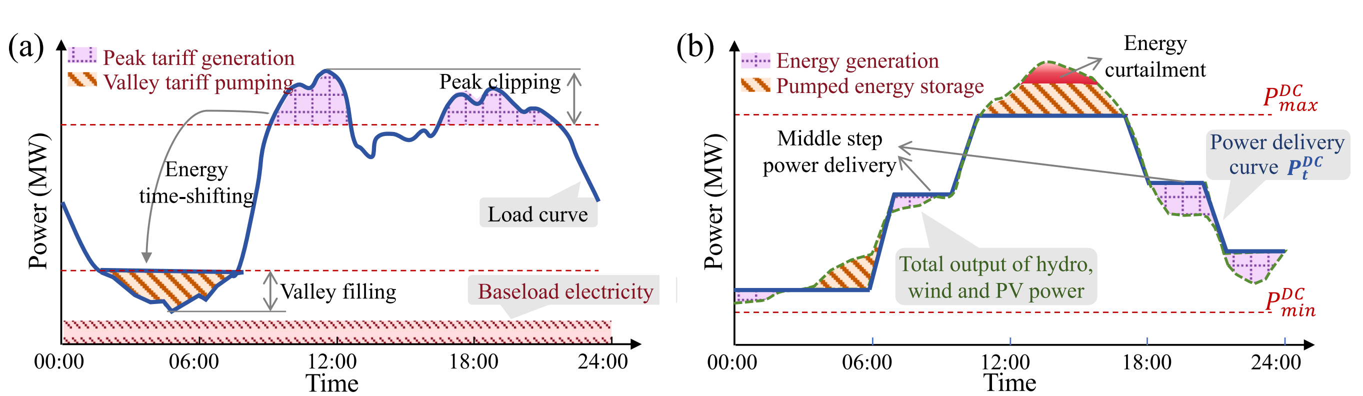


Fig. 2. Schematic of daily operating processes of the pumped hydro storage under different operating modes. (a) Conventional operating mode, and (b) Integrated delivery operating Mode.

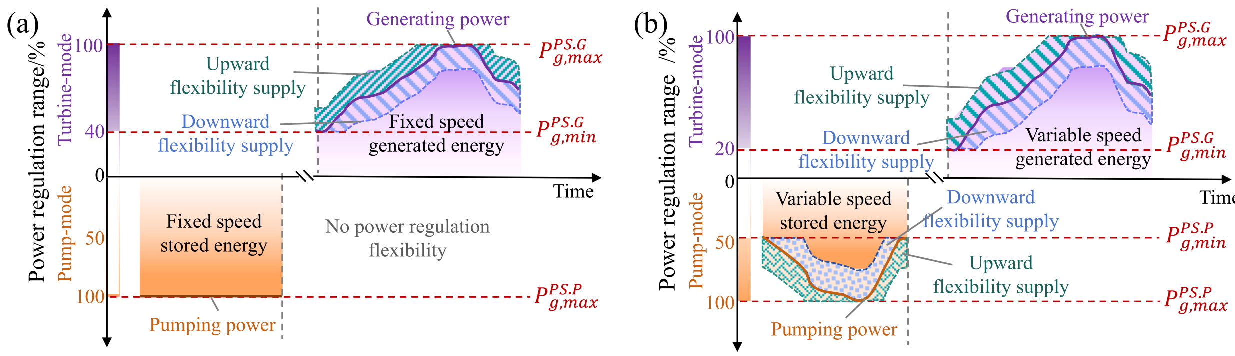


Fig. 3. Flexibility adjustment range of fixed-speed and variable-speed pumped hydro storage (PHS) units in turbine and pump Modes. (a) Fix-speed PHS, and (b) Variable-speed PHS.

reservoir's water volume at time t, its maximum, and minimum operational limits, respectively. $\xi PS\ G$ and $\xi PS\ P$ denote the hydraulic conversion coefficients of unit g for flow regulation in generation and pumping modes, respectively.

2.2.2. Frequency response of pumped hydro storage units

FS-PHS units are directly connected to the grid through synchronous motors. Their frequency response characteristics constrained by both the operational state of the units and the mechanical inertia of the motors. When the system frequency deviation exceeds a predetermined deadband, FS-PHS units rapidly support the grid by discharging stored rotor kinetic energy. However, under pumping conditions, the inherent synchronous speed locking restricts these units to only limited inertial support. As a result, FS-PHS units are typically unable to participate in primary frequency regulation (PFR) during pumping mode and instead contribute to frequency regulation mainly in generation mode. In this mode, their response mechanism closely mirrors that of conventional hydroelectric generators, where frequency adjustments are achieved by modulating the guide vane opening via a mechanical-hydraulic governor. Fig. B.1 illustrates the corresponding frequency response model, which encapsulates three principal components: synchronous inertia, turbine dynamics, and governor control.

In contrast, VS-PHS units employ power electronic converters with advanced control strategies to decouple power output from grid frequency. This decoupling enables these units to provide frequency support in both generation and pumping modes, thereby enhancing the overall system stability. In terms of inertial response, VS-PHS units simulate the inertial characteristics of synchronous generators through virtual inertia control [30]. When grid frequency decreases, these units rapidly adjust their output power, delivering inertial support comparable to that of traditional synchronous machines. For frequency regulation, VS-PHS units take advantage of variable frequency control technology, which enables bidirectional regulation and allows for precise adjustments to frequency fluctuations. Specifically, in the event of a frequency drop, the unit increases generation power or reduces pumping power, and conversely, when the frequency rises, it decreases generation power or increases pumping load. VS-PHS units typically implement a Proportional-Integral (PI) control strategy for frequency regulation. The PI controller was chosen for its well-established advantages in power system applications. Its straightforward design allows for rapid implementation and fine-tuning, while the proportional component ($KP\ PS$) offers immediate corrective action in response to frequency deviations, and the integral component ($KI\ PS$) effectively eliminates steady-state errors to ensure long-term stability. These characteristics, along with extensive engineering validation in related studies [31], render the PI control approach particularly well-suited for managing the dynamic responses of PHS units. Fig. 4 presents a schematic diagram of the dynamic frequency response module for PHS units employing the PI control strategy [32]. In this diagram, f_{ref} is the reference frequency, f denotes the unit frequency, Δf is the deviation of frequency, s denotes the differential operator, and e_p denotes the adjustment coefficient.

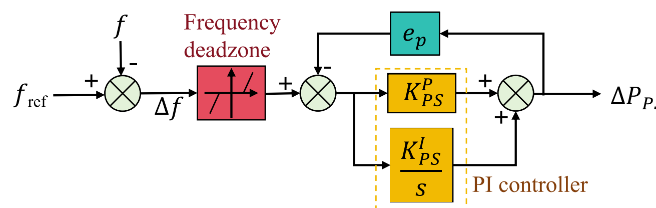


Fig. 4. Structural block diagram of primary frequency regulation control module for variable-speed PHS units.

When the system frequency deviates from the reference value, the unit utilizes a PI controller in conjunction with the e_p, to convert the frequency deviation (Δf) into an active power deviation command (ΔP_{PS}), thereby modulating its output power accordingly. Its transfer function can be expressed as [33]:

$$G_{PS}(s) = \frac{\Delta P_{PS}(s)}{\Delta f(s)} = \frac{K^P_{PS} + K^I_{PS}/s}{1 + e_p\left(K^P_{PS} + K^I_{PS}/s\right)} \tag{7}$$

Despite the marked disparities in dynamic characteristics between fixed-speed and variable-speed units, both types are inherently subject to physical constraints that limit their frequency regulation capacity. Consequently, the PFR process can be approximated by the piecewise linear function as illustrated in Fig. 5 [34], and expressed in Eq. (8).

$$PFR^{PS}_{g,t} = \begin{cases} 0, & t \le t_{db} \\ \dfrac{\overline{PFR}^{PS}_g}{T_d}(t - t_{db}), & t_{db} \le t \le t_{db} + T_D \\ \overline{PFR}^{PS}_g = k^{PS}_{f.g} \cdot Cap^{PS}_g, & t_{db} + T_D \le t \end{cases} \tag{8}$$

where, $\overline{PFR}^{PS}_g$ is the upper limit of PFR capacity for unit g, $k^{PS}_{f.g}$ is the PFR

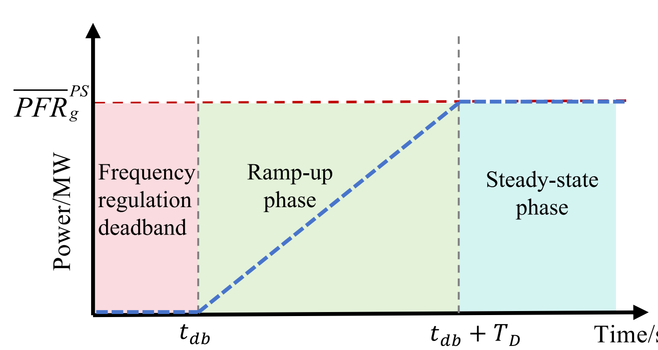


Fig. 5. Piecewise linear function of frequency regulation capacity.

capability coefficient, t_{dB} indicates the moment when the frequency exceeds the deadband, T_D represents the delivery time of frequency regulation capacity, and Cap_g^{PS} refers to the unit g rated capacity.

The PFR reserve capacity provided by the participating PHS units is constrained by their available headroom. Hence, the PFR reserve supply of the PHS unit g at time t can be expressed as:

$$\begin{cases} R_{PFR,g,t}^{PS.G} = \min\left[PFR_{g,t}^{PS}, u_{g,t}^{PS.G} \cdot P_{g,\max}^{PS.G} - P_{g,t}^{PS.G}\right] \\ R_{PFR,g,t}^{PS.P} = \min\left[PFR_{g,t}^{PS}, P_{g,t}^{PS.P} - u_{g,t}^{PS.P} \cdot P_{g,\min}^{PS.P}\right] \end{cases} \tag{9}$$

where $R_{PFR,g,t}^{PS.G}$ and $R_{PFR,g,t}^{PS.P}$ denote the PFR reserve capacities provided under pumping and generating operating conditions, respectively.

2.3. Flexibility balancing constraints of the integrated delivery system

The flexibility balancing constraints of CHIDS are realized via a synergistic, multi-timescale mechanism encompassing steady-state, transient, and quasi-steady-state phases. In the steady-state phase (minutes to hours), supply-demand equilibrium is maintained through the optimization of generation outputs and reserve capacity allocation, thereby maximizing transmission corridor utilization. During the transient phase (seconds to minutes), frequency instability due to abrupt disturbances is mitigated by inertial response and PFR, with safety boundaries validated via offline dynamic frequency response simulations. In the quasi-steady-state phase (minutes), secondary frequency regulation (SFR) eliminates residual frequency deviations to restore steady-state operation.

2.3.1. Power regulation flexibility balancing constraints

(1) Power regulation flexibility requirements

The demand for power regulation flexibility is predominantly driven by the inherent uncertainties associated with wind and PV power generation. To quantitatively assess this demand, this study utilizes a prediction model based on confidence intervals, employing the Bootstrap method as outlined in [35]. By applying the designed confidence level α, the fluctuation ranges of wind and PV power are estimated using Eq. (10).

$$\begin{cases} \underline{P}_{\alpha,t}^{WT} \le P_{f,t}^{WT} \le \overline{P}_{\alpha,t}^{WT} \\ \underline{P}_{\alpha,t}^{PV} \le P_{f,t}^{PV} \le \overline{P}_{\alpha,t}^{PV} \end{cases} \tag{10}$$

where $\underline{P}_{\alpha,t}^{WT}$ and $\overline{P}_{\alpha,t}^{WT}$ denote the lower and upper limits of wind power output at time t, $\underline{P}_{\alpha,t}^{PV}$ and $\overline{P}_{\alpha,t}^{PV}$ represent the corresponding limits for PV power output, respectively.

Based on this, the system's flexibility requirements are categorized into upward and downward demands. Upward demand involves the rapid increase of power generation from controllable sources when wind and PV outputs fall below predicted levels, thus maintaining supply-demand equilibrium and reducing the risk of shortages. Conversely, downward demand requires the reduction of surplus power generation when outputs exceed foretasted levels, thus preventing power surpluses that could cause frequency fluctuations and threaten grid stability. The specific requirements for upward and downward power regulation flexibility, along with their respective quantification methods, are presented as follows:

$$\begin{cases} R_{up,t}^{req} = \max\left(0, \left(P_{f,t}^{WT} - \underline{P}_{\alpha,t}^{WT}\right) + \left(P_{f,t}^{PV} - \underline{P}_{\alpha,t}^{PV}\right)\right) \\ R_{dn,t}^{req} = \max\left(0, \left(\overline{P}_{\alpha,t}^{WT} - P_{f,t}^{WT}\right) + \left(\overline{P}_{\alpha,t}^{PV} - P_{f,t}^{PV}\right)\right) \end{cases} \tag{11}$$

(2) Power regulation flexibility supply

In the CHIDS, power regulation flexibility is primarily provided by PHS and hydropower units. The upward and downward flexibility supply of PHS stations is detailed in Eqs. (5) and (6), while the flexibility supply of hydropower stations is described in Eq. (12).

$$\begin{cases} R_{up,t}^{HP} = \sum\limits_{i \in N_{HP}} \sum\limits_{g \in N_{HP,i}} \min\left(\Delta P_{ramp,i,g}^{HP} \Delta t, u_{i,g,t}^{HP} P_{i,g,\max}^{HP} - P_{i,g,t}^{HP}, \dfrac{V_{i,t}^{HP} - V_{i,\min}^{HP}}{\xi_{HP,i} \Delta t}\right) \\ R_{dn,t}^{HP} = \sum\limits_{i \in N_{HP}} \sum\limits_{g \in N_{HP,i}} \min\left(\Delta P_{ramp,i,g}^{HP} \Delta t, P_{i,g,t}^{HP} - u_{i,g,t}^{HP} \cdot P_{i,g,\min}^{HP}\right) \end{cases} \tag{12}$$

where N_{HP} represents the number of hydropower stations, $\xi_{HP,i}$ is the hydraulic conversion coefficients of station i, $R_{up,t}^{HP}$ and $R_{dn,t}^{HP}$ denote the upward and downward reserve capacities provided by the hydropower at time period t, respectively; $\Delta P_{ramp,i,g}^{HP}$ indicates the ramping rate of unit g within station i.

(3) Power regulation flexibility supply and demand balance

Furthermore, the CHIDS's power regulation flexibility necessitates that the supply capacity meets or exceeds actual regulation demand to sustain supply-demand balance amid fluctuations in wind and PV output. The constraints are as follows:

$$\begin{cases} R_{up,t}^{HP} + R_{up,t}^{PS} \ge R_{up,t}^{req} \\ R_{dn,t}^{HP} + R_{dn,t}^{PS} \ge R_{dn,t}^{req} \end{cases} \tag{13}$$

2.3.2. Transient frequency response flexibility balancing constraints

(1) Transient frequency response flexibility requirements

Transient frequency flexibility is a critical indicator for evaluating a power system's ability to respond to sudden disturbances. Such disturbances may stem from the variability of VRE sources, generator disconnections, or transmission line faults. After a disturbance, the frequency response may either increase or decrease, contingent on whether there is an excess or deficit in system power. In integrated delivery systems with a high proportion of VRE, the low inertia characteristics present a heightened risk of frequency decreases. Fig. 6 illustrates the dynamic frequency response process following a sudden reduction in wind or PV output, encompassing three stages: inertial response, primary frequency control, and secondary frequency control [36].

During the inertial response phase ($t_0 \sim t_d$), the rotational kinetic energy of generators counteracts the rapid decline in frequency during the initial seconds of a power disturbance, thus providing essential energy buffering for the system. Primary frequency response ($t_d \sim t_{ss}$) is triggered when the frequency deviation surpasses the deadband,

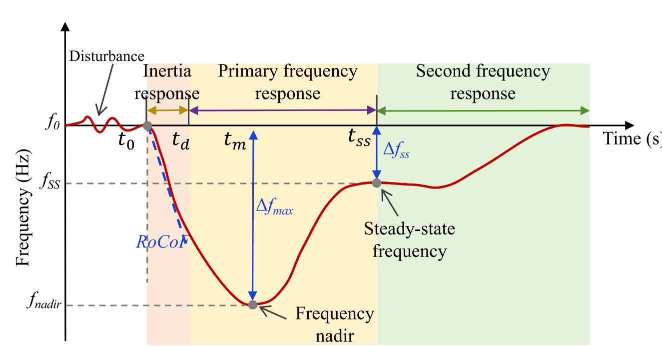


Fig. 6. The dynamic frequency response of the integrated delivery system after disturbance.

enabling the system to promptly adjust the output of controllable generators, thereby reducing the rate of frequency decline and stabilizing it within a quasi-steady state. SFR further fine-tunes the generator output via an automatic generation control system, facilitating the gradual restoration of frequency to its pre-disturbance steady-state value. The dynamic evolution of system frequency can generally be represented by the classical swing equation:

$$\frac{2H_{sys}}{f_0}\cdot\frac{d\Delta f(t)}{dt}+k_d\cdot\Delta f(t)=\Delta P_m(t)-\Delta P_d(t) \tag{14}$$

where f_0 represents the nominal frequency of the system, H_{sys} is the system inertia, k_{d} represents the load damping rate, $\Delta P_m(t)$ and $\Delta P_d(t)$ indicate the total PFR power provided by all units and power imbalance at time t, respectively.

Transient frequency safety is commonly quantified using three metrics: the rate of change of frequency (*RoCoF*), the maximum frequency deviation (*MFD*), and the quasi-steady-state frequency deviation (*QSFD*).

The *RoCoF* is defined as the instantaneous frequency gradient during the initial disturbance period, i.e. $RoCoF = d\Delta f(t)/dt$. When the *RoCoF* exceeds the prescribed limit (typically $RoCoF_{\max} = 0.5$ Hz/s). Eq. (15) is then employed to ensure that the system maintains a minimum inertia reserve.

$$H_{sys,t}^{\min}=\frac{\Delta P_{d,t}\cdot f_0}{2RoCoF_{\max}} \tag{15}$$

The *MFD* represents the deviation at the frequency nadir, denoted as Δf_{nadir}. If Δf_{nadir} exceeds the established safety limit Δf_{nadir}^{max}, it may activate low-frequency load-shedding devices, increasing the risk of widespread blackouts. Hence, the constraint for the extreme value of frequency deviation is:

$$|\Delta f_{nadir}|\le\Delta f_{naidr}^{\max} \tag{16}$$

The *QSFD* denotes the frequency offset observed after PFR and before the initiation of SFR, and it must satisfy the constraint presented in Eq. (17).

$$|\Delta f_{ss}|\le\Delta f_{ss}^{\max} \tag{17}$$

(2) Modeling of Flexibility Provision

The provision of transient frequency safety flexibility quantifies the power system's capacity to maintain frequency stability under abrupt power disturbances via a multi-source coordinated mechanism. This capability is primarily determined by the CHIDS's equivalent inertia reserve, rapid-response PFR margin, and the dynamic complementary interactions among heterogeneous generation units. To assess flexibility provision quantitatively, a multi-machine dynamic frequency response model was developed, as illustrated in Fig. 7. This model is based on the generator frequency response model presented in Section 2.2.2, noting that the transfer function for FS-PHS under generation conditions aligns with that of hydroelectric units, as depicted in Appendix B.

The inertia of the CHIDS, serving as the primary safeguard for frequency stability, is jointly determined by the operating status, rated capacity, and inertia characteristics of each unit. At time t, the total inertia of the system $H_{sys,t}$, can be expressed as:

$$H_{sys,t}=\sum_{g\in N_{HP}}u_{g,t}^{HP}P_{g,\max}^{HP}H_g^{HP}+\sum_{g\in N_{HP}}\left(u_{g,t}^{PS,G}+u_{g,t}^{PS,P}\right)P_{g,\max}^{PS}H_g^{HP} \tag{18}$$

where $P_{i,max}^{HP}$ and $H_{i,g}^{HP}$ are the maximum output and inertia of hydro unit g, respectively.

The analytical expression for the *MFD* of the system following a disturbance ΔP_d is derived from a comprehensive dynamic frequency response model, as indicated in Eq. (19) [37]. Detailed derivations and expressions for other parameters are provided in Appendix C.

$$\Delta f_{ndair,t}=\frac{f_0\cdot\Delta P_{d,t}}{D+R_{T,t}}\left(1+\sqrt{\frac{T_{R,t}(R_{T,t}-F_{H,t})}{2H_{sys,t}}}e^{-\xi\omega_n t_{nadir}}\right) \tag{19}$$

where ω_n is the natural frequency, ξ is the damping ratio, t_{naidr} is the time instance of frequency nadir, $F_{H,t}$, $T_{R,t}$ and $R_{T,t}$ are the fraction of total power generated by the turbine, the reheat time constant, and the droop gain of the equivalent aggregate system, respectively.

As the system frequency approaches stability in the vicinity of steady state, the terminal value theorem can be employed to derive the expression for the *QSFD* [38].

$$\Delta f_{ss}=\lim_{s\to 0}s\Delta f=\frac{f_0\cdot\Delta P_d}{D+R_{T,t}} \tag{20}$$

The PFR reserve capacity offered by the hydropower units involved in the frequency regulation of the CHIDS is expressed as follows:

$$R_{PFR,i,g,t}^{HP}=\min\left[k_{f,i,g}^{HP}\cdot Cap_{i,g}^{HP},u_{i,g,t}^{HP}\cdot P_{i,g,\max}^{HP}-P_{i,g,t}^{HP}\right] \tag{23}$$

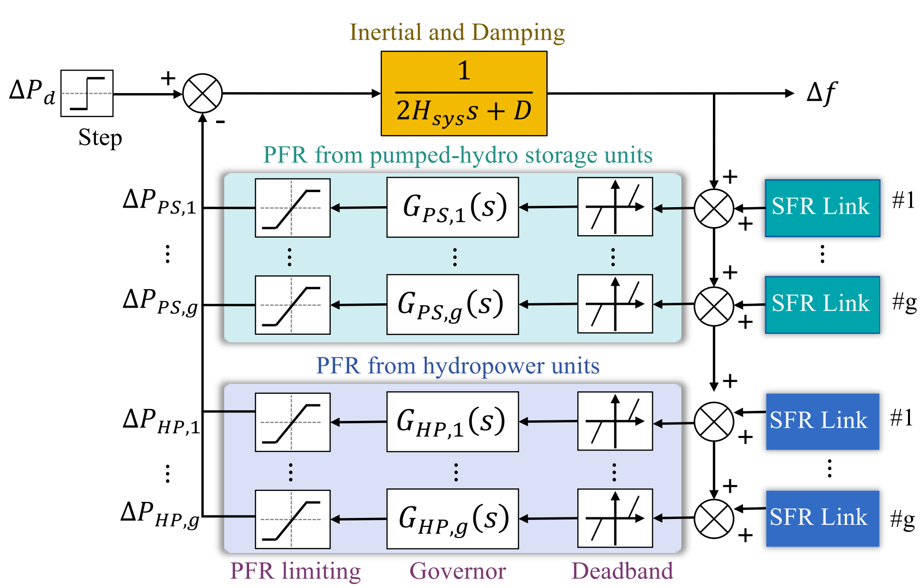


Fig. 7. Dynamic frequency response model for the integrated delivery system.

where $k_{f,i,g}^{HP}$ represents the PFR capacity coefficient for hydropower unit g in station i.

By integrating the PFR reserve from the PHS and hydropower, the overall PFR flexibility supply of the CHIDS can be articulated as follows:

$$R_{PFR,t}^{sys} = \sum_{i}^{N_{HP}} \sum_{g}^{N_{HP,i}} R_{PFR,i,g,t}^{HP} + \sum_{g}^{N_{PS}} R_{PFR,g,t}^{PS} \tag{24}$$

(3) Transient frequency security flexibility supply-demand balance constraints

The system must first meet the minimum inertia requirement specified in Eq. (25), ensuring that the *RoCOF* remains within a safe range to prevent instability.

$$H_{sys,t} \geq H_{sys,t}^{\min} \tag{25}$$

At the time t_{naidr}, when the PFR power equals the system's power imbalance, the system reaches its *MFD*. By integrating Eqs. (16) and (19), conditions for ensuring system frequency safety, as outlined in Eq. (26), can be derived.

$$\frac{|\Delta P_{d,t}| - R_{PFR,t}^{sys}}{D + R_{T,t}} \left(1 + \sqrt{\frac{T_R (R_{T,t} - F_{H,t})}{2H_{sys,t}}} e^{-\xi \omega_n t_{nadir}} \right) \leq \frac{\Delta f_{naidr}^{\max}}{f_0} \tag{26}$$

Additionally, to maintain frequency deviation controllability post-regulation, Eqs. (17) and (20) must be combined to establish constraints for the *QSFD*, detailed in Eq. (27).

$$\frac{|\Delta P_{d,t}| - R_{PFR,t}^{sys}}{D + R_{T,t}} \leq \frac{\Delta f_{ss}^{\max}}{f_0} \tag{27}$$

It should be noted that Eq. (27) ensures that the *QSFD* remains within the acceptable range but does not fully compensate for the power deficit. To achieve complete frequency restoration, SFR must be introduced.

2.3.3. Quasi-steady state frequency recovery constraint

The quasi-steady state defines a transitional phase where transient frequency deviations are temporarily stabilized, enabling SFR to restore the system to its nominal steady-state frequency [39].This phase is physically governed by the decoupling of timescales: fast dynamics (inertia, PFR) suppress immediate instability, while slower SFR reserves address residual imbalances. To ensure *QSFD* recovery, the aggregate SFR capacity must exceed the post-PFR power imbalance, as formulated in Eq. (28).

$$\sum_{i \in N_{HP}} \sum_{g \in N_{HP,i}} \left(R_{PFR,i,g,t}^{HP} + R_{SFR,i,g,t}^{HP} \right) + \sum_{g \in N_{PS}} \left(R_{PFR,g,t}^{PS} + R_{SFR,g,t}^{PS} \right) \geq |\Delta P_{d,t}| \tag{28}$$

2.4. Multi-timescale optimization scheduling model

This section introduces a multi-timescale optimization model for the CHIDS, using a hierarchical framework to optimize day-ahead economic performance and intraday security, as depicted in Fig. 8. To reconcile transient frequency stability with steady-state economic objectives, an offline-online coordination mechanism is established. In the offline phase, leveraging the dynamic frequency response model illustrated in Fig. 7, second-level time-domain simulations ($\Delta t = 0.1$ s) are performed to quantify the security boundary parameters under extreme power disturbance scenarios, including the minimum system inertia requirement, PFR capacity, and quasi-steady-state recovery capacity. These boundaries define the permissible operational regions for subsequent scheduling processes. During the online intraday rolling optimization ($\Delta t^{ID} = 15$ min), a sliding time-window averaging algorithm and confidence interval correction transform dynamic security parameters into 15-min constraints. These constraints are integrated into intraday dispatch model, thereby circumventing the computational burden of real-time dynamic simulations during iterative optimization while preserving frequency security guarantees. In the day-ahead stage ($\Delta t^{DA} = 1$ h), the power delivery schedule is optimized to maximize economic efficiency, whereas the intraday stage enforces transient security constraints derived from offline boundaries, thereby achieving an optimal trade-off between economic efficiency and frequency stability.

2.4.1. Uncertainty modeling of wind and PV output

(1) Day-Ahead wind and PV output scenario construction

In day-ahead scheduling, the concept of "scenario combinations" is introduced to construct a representative scenario set that captures the output characteristics of wind and PV power, improving computational efficiency while preserving critical information. The specific steps are as follows:

Step 1: Seasonal feature extraction. Based on the seasonal patterns of wind and PV generation, historical and forecast data are classified

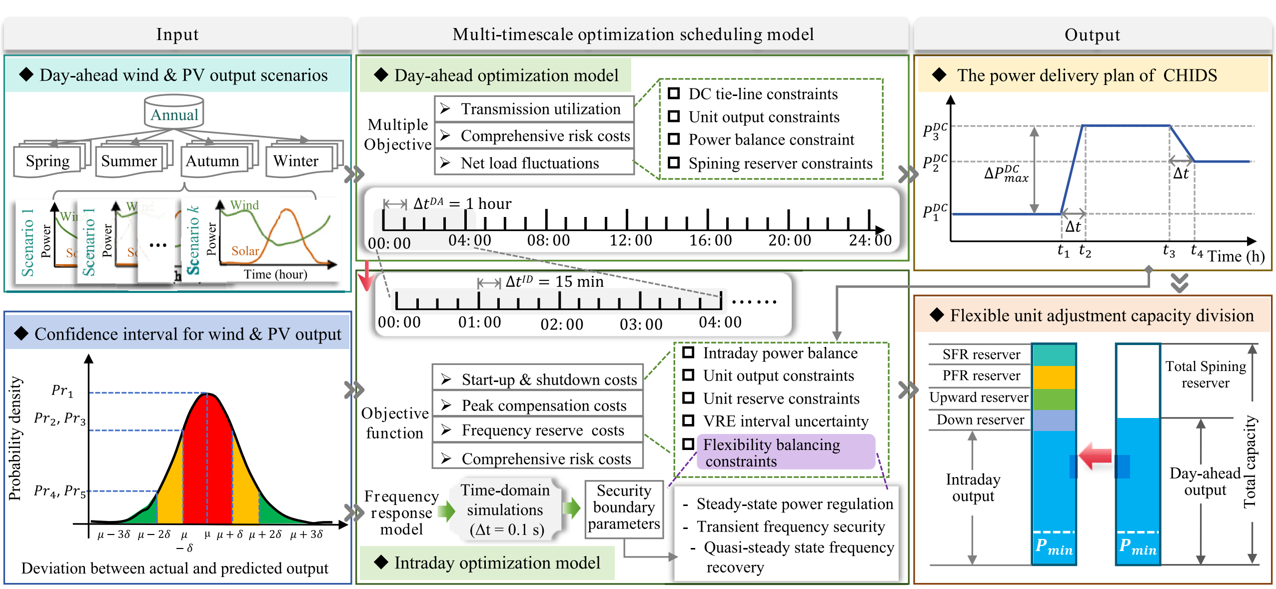


Fig. 8. Multi-time scale optimization framework for the integrated delivery system.

into four seasons. For each season, statistical techniques are employed to extract critical meteorological and output features, such as wind speed, solar irradiance, temperature, and the mean, standard deviation, maximum, and minimum output values.

Step 2: Multidimensional scenario clustering. The K-means clustering algorithm is employed to analyze wind and PV output data across various seasons [40]. Each season's scenario feature vector is defined as $\mathbf{x}_t = \{PWTf, PPVf\}$, where $PWTf$ and $PPVf$ denote the outputs of wind and PV, respectively. Then, Euclidean distance is employed to quantify the similarity between scenarios, with the clustering objective focused on minimizing intra-cluster distances, as defined Eq. (29).

$$\min \sum_{k\in N_{ck}} \sum_{x\in C_k} \|x-\mu_k\|^2 \tag{29}$$

where μ_k is the centroid of the k^{th} cluster, C_k denotes the k^{th} cluster, N_{ck} is the total number of clusters.

Step 3: Scenario probability calculation. To enhance the applicability of scenario forecasting, the occurrence probability of each typical scenario is derived from the frequency distribution of historical data. Assuming historical data accurately reflects future output conditions, the probability of scenario S_j is defined as its historical frequency $P(S_j)$, computed as follows:

$$P(S_j) = n_j / N_{Total} \tag{30}$$

where, n_j is the frequency of occurrence for scenario S_j, N_{Total} is the total number of all scenarios.

Step 4: Construction of the typical output scenario set. The scenario set $\mathbf{S} = \{S_1, S_2, \dots, S_K\}$ encompasses typical combined output scenarios of wind and PV generation for each season, along with their occurrence probabilities. The final scenario set is represented as::

$$S = \left\{ \left(k, p_k, P_{f,k}^{WT}(t), P_{f,k}^{PV}(t) \right) \right\}, \quad t\in[1,24] \tag{31}$$

where p_k is the probability of scenario k.

(2) Confidence intervals for intraday wind and PV output forecasting

Based on a collection of typical output scenarios from day-ahead forecasts, a confidence interval for the intra-day wind and PV output predictions at a 15-min resolution is constructed using the Bootstrap method. The main steps are as follows:

Step 1: Temporal resolution conversion. Hourly wind and PV output forecast data are converted to a 15-min resolution using spline interpolation, yielding more refined output predictions. The interpolated wind and PV outputs are defined as:

$$P_{f,k}^{WT(ID)}(t) = BS_{WT}\left(P_{f,i}^{WT}(t)\right), \quad P_{f,k}^{WT(ID)}(t) = BS_{WT}\left(P_{f,i}^{WT}(t)\right) \tag{32}$$

where BS_{WT} and BS_{PV} are the interpolation functions.

Step 2: Prediction error modeling. The intraday prediction errors for wind and PV output are assumed to follow normal distributions $N(0, \sigma 2\ \mathrm{WT})$ and $N(0, \sigma 2\ \mathrm{PV})$, respectively. Standard deviations σ_{WT} and σ_{PV} are estimated from historical data, with the assumption that errors are independent and identically distributed across 15-min intervals.

Step 3: Constructing confidence intervals. Based on the output predictions $P_{f,k,t}^{WT(ID)}$ and $P_{f,k,t}^{PV(ID)}$, the Bootstrap method is employed for resampling to generate Bootstrap samples. Calculating the mean and standard deviation of the predicted outputs for each sample. The mean and standard deviation of predicted outputs are computed for each sample. The confidence interval at a specified level α is then defined as follows:

$$\begin{cases} CI_{WT}(t) = \left[P_{f,k}^{WT(ID)}(t) - z_{\alpha/2}\sigma_{WT}, P_{f,k}^{WT(ID)}(t) + z_{\alpha/2}\sigma_{WT}\right] \\ CI_{PV}(t) = \left[P_{f,k}^{PV(ID)}(t) - z_{\alpha/2}\sigma_{PV}, P_{f,k}^{PV(ID)}(t) + z_{\alpha/2}\sigma_{PV}\right] \end{cases} \tag{33}$$

where $z_{\alpha/2}$ is the quantile of the standard normal distribution, with $z_{0.025} \approx 1.96$ for a 95 % confidence interval. This method establishes upper and lower limits for wind and PV output predictions at time t.

2.4.2. Day-ahead power delivery optimization model

The day-ahead power delivery optimization model aims to formulate an optimal power delivery strategy based on typical forecast scenarios for wind and photovoltaic output, along with the load demand at the receiving end, with a one-hour temporal resolution. The optimization objectives are centered on three key aspects: (1) maximizing the utilization of the UHVDC transmission channel to enhance the economic efficiency of power delivery; (2) minimizing the risks of power curtailment and shortages, thereby reducing the uncertainty costs associated with system operation; and (3) addressing the load regulation requirements of the receiving-end grid while mitigating fluctuations in its net load.

$$\max \Pi_1 = \sum_{k\in N_k} p_k \cdot \left(\sum_{t\in N_T} P_{k,t}^{DC} \Big/ P_{\max}^{DC} \cdot N_T \right) \tag{34}$$

$$\max \Pi_2 = -\sum_{k\in N_k} p_k \cdot \sum_{t\in N_T} \left(c_{spill} P_{k,t}^{spill} + c_{cur} P_{k,t}^{cur} + c_{loss} P_{k,t}^{loss} \right) \tag{35}$$

$$\max \Pi_3 = -\sum_{k\in N_k} p_k \cdot \sqrt{\frac{1}{T}\sum_{t\in N_T}\left[P_{k,t}^{Load} - \left(\sum_{t\in N_T} |P_{k,t}^{Load} - P_{k,t}^{DC}| / N_T \right) \right]^2} \tag{36}$$

where N_k is the number of representative scenarios, $P_{k,t}^{DC}$ indicates the delivery power during time period t under scenario k, P_{max}^{DC} is the maximum transmission capacity, c_{cur}, c_{spill}, and c_{loss} are the penalty costs for curtailment, spillage, and power shortage, respectively. $P_{k,t}^{spill}$ reflects the spillage from hydropower station i, $P_{k,t}^{cur}$ and $P_{k,t}^{loss}$ are the curtailment and power shortage at time t, respectively. $P_{k,t}^{Load}$ reflects the load demand at the receiving-end under scenario k for time period t.

The constraints for the day-ahead scheduling model are as follows:

(1) Transmission interconnection operation constraints. This study reformulates the operation model of the direct current interconnection line into a power transmission framework, following the methodology in [41]. The model assumes that transmitted power adheres to a stepped output pattern, constrained by interconnection capacity, adjustment magnitude, and minimum stability duration. This can be expressed as:

$$P_{k,t}^{DC} = \left\{ P_{k,t}^1, \cdots, P_{k,\tau_1-1}^1, P_{k,\tau_1}^2, \cdots, P_{k,\tau_2-1}^2, \cdots, P_{k,\tau_s-1}^s, \cdots, P_{k,N_T}^S \right\} \tag{37}$$

$$P_{\min}^{DC} \le P_{k,t}^{DC} \le P_{\max}^{DC} \tag{38}$$

$$|P_{k,t}^{DC} - P_{k,t-1}^{DC}| \le \Delta P_{\max}^{DC} \tag{39}$$

$$\tau_s - \tau_{s-1} \ge \tau_d \tag{40}$$

where S is the number of power delivery adjustments, $P_{k,t}^{S}$ is the delivery power during the s^{th} stable operating period of CHIDS under scenario k, τ_s denotes the time of power change, P_{min}^{DC} indicates the minimum delivery power limit, $\Delta PDC\ max$ denotes the maximum fluctuation,

τ_d specifies the minimum duration for which the interconnection line must remain stable following a power change.

(2) Conventional operating constraints for units. The operational constraints for cascade hydropower are detailed in Eqs. (A.1)–(A.11), the output constraints for wind and PV generation are specified in Eqs. (A.12) and (A.13). The constraints for PHS are outlined in Eqs. (A.14)–(A.24).

(3) Power balance constraint.

$$\sum_{i\in N_{HP}}\sum_{g\in N_{HP,g}} P^{HP}_{k,i,g,t}+P^{WT}_{k,t}+P^{PV}_{k,t}+\sum_{g\in N_{PS}}\left(P^{PS.G}_{k,g,t}-P^{PS.P}_{k,g,t}\right)+P^{loss}_{k,t}=P^{DC}_{k,t} \tag{41}$$

where $P^{WT}_{k,t}$ and $P^{PV}_{k,t}$ are the actual output of wind and PV at time t under scenario k, respectively.

(4) Power shortage constraint. To ensure reliable power delivery, the deviation between CHIDS's output and required demand must be limited within a specified range.

$$0\le P^{loss}_{k,t}\le \varepsilon_{DC} \tag{42}$$

where ε_{DC} represents the precision control index of the CHIDS's output.

(5) VRE curtailment constraint.

$$\begin{cases} 0\le \sum_{t\in N_T} P^{cur}_{k,t}\Big/\sum_{t\in N_T}\left(P^{WT}_{f,k,t}+P^{PV}_{f,k,t}\right)\le \mu^{\max}_{VRE} \\ P^{cur}_{k,t}=\left(P^{WT}_{f,k,t}-P^{WT}_{k,t}\right)+\left(P^{PV}_{f,k,t}-P^{PV}_{k,t}\right) \end{cases} \tag{43}$$

where μ^{max}_{VRE} represents the upper limit of the VRE curtailment rate.

(6) Spinning reserve constraint: In the day-ahead scheduling, hydropower and PHS units must provide sufficient spinning reserve capacity to compensate forecast errors in wind and PV output. Positive reserves address overestimations, while negative reserves counteract underestimations.

$$\begin{cases} R^{HP}_{up,k,t}+R^{PS}_{up,k,t}\ge SR_{up,t} \\ R^{HP}_{dn,k,t}+R^{PS}_{dn,k,t}\ge SR_{dn,t} \end{cases} \tag{44}$$

where $SR_{up,t}$ and $SR_{dn,t}$ represent the upward and downward spinning reserve requirements of the CHIDS, respectively, contingent upon the forecast errors in wind and solar output.

2.4.3. Intraday economic dispatch model

The intraday scheduling model aims to optimize flexibility resource allocation while addressing the uncertainty in ultra-short-term forecasts for wind and photovoltaic output. This ensures frequency stability and provides sufficient ramping and frequency regulation margins for future periods. The goal is to align actual intraday output with planned interconnection delivery, minimizing the system's total operational cost over the scheduling period. The optimization objective for the intraday phase incorporates the following cost components:

(1) Total start-up cost of flexible power sources in the CHIDS:

$$\begin{aligned} F^{on}_{k,t}=&\sum_{i\in N_{HP}}\sum_{g\in N_{HP,i}} c^{HP}_{on,i}\cdot u^{HP(ID)}_{k,i,g,t}\cdot\left(1-u^{HP(ID)}_{k,i,g,t-1}\right)+ \\ &\sum_{g\in N_{PS}} c^{PS}_{on}\cdot\left[u^{PS.G(ID)}_{k,g,t}\cdot\left(1-u^{PS.G(ID)}_{k,g,t-1}\right)+u^{PS.P(ID)}_{k,g,t}\cdot\left(1-u^{PS.P(ID)}_{k,g,t-1}\right)\right] \end{aligned} \tag{45}$$

where $F^{on}_{k,t}$ is the start-up cost of the CHIDS at time t under scenario k. $c^{HP}_{on,i}$ and c^{PS}_{on} are the start-up costs of a single hydropower and pumped storage unit, respectively.

(2) Peak shaving reserve opportunity cost (*FPRC k,t*):

$$F^{PRC}_{k,t}=c^{PS}_{peak}\cdot\left(R^{PS(ID)}_{up,k,t}+R^{PS(ID)}_{dn,k,t}\right)+c^{PS}_{peak}\cdot\left(R^{HP(ID)}_{up,k,t}+R^{HP(ID)}_{dn,k,t}\right) \tag{46}$$

where *CPS peak* and *CHP peak* are the peak shaving reserve costs for PHS and hydropower unit, respectively.

(3) Frequency regulation reserve opportunity cost (*FFRC k,t*):

$$\begin{aligned} F^{FRC}_{k,t}=&\sum_{i\in N_{HP}}\sum_{g\in N_{HP,i}} c^{HP}_{PFR}\cdot R^{HP(ID)}_{PFR,k,i,g,t}+\sum_{g\in N_{PS}} c^{PS}_{PFR}\cdot R^{PS(ID)}_{PFR,k,g,t}+ \\ &\sum_{i\in N_{HP}}\sum_{g\in N_{HP,i}} c^{HP}_{SFR}\cdot R^{HP(ID)}_{SFR,k,i,g,t}+\sum_{g\in N_{PS}} c^{PS}_{SFR}\cdot R^{PS(ID)}_{SFR,k,g,t} \end{aligned} \tag{47}$$

where *CPS PFR* and *CHP PFR* denote the PFR reserve costs for PHS unit and hydropower unit, respectively, while *CPS SFR* and *CHP SFR* represent the SFR reserve costs for these units.

(4) Total risk cost of power curtailment and load shedding (*FRisk k,t*):

$$F^{Risk}_{k,t}=c_{spill}\cdot P^{spill(ID)}_{k,t}+c_{cur}\cdot P^{cur(ID)}_{k,t}+c_{loss}\cdot P^{loss(ID)}_{k,t} \tag{48}$$

Based on this consideration, the intraday optimization objective function is as follows:

$$\min F_{sys}=\sum_{k\in N_k}\sum_{t\in N^{ID}_T} p_k\cdot\left(F^{on}_{k,t}+F^{PRC}_{k,t}+F^{FRC}_{k,t}+F^{Risk}_{k,t}\right)\cdot\Delta t^{(ID)} \tag{49}$$

The constraints of the intraday scheduling model, in addition to the conventional operational constraints of each unit and spinning reserve constraint the system, include the following:

(1) Intraday power balance constraint.

$$\begin{aligned} &\sum_{i\in N_{HP}}\sum_{g\in N_{HP,g}} P^{HP(ID)}_{k,i,g,t}+P^{WT(ID)}_{k,t}+P^{PV(ID)}_{k,t}+\sum_{g\in N_{PS}}\left(P^{PS.G(ID)}_{k,g,t}-P^{PS.P(ID)}_{k,g,t}\right)+P^{loss(ID)}_{k,t} \\ &=P^{DC}_{k,t} \end{aligned} \tag{50}$$

(2) Flexibility unit reserve constraints. The allocation diagram for intraday flexibility unit reserve capacity is presented in Fig. 8. Each flexibility resource must reserve adequate peak shaving and frequency regulation capacity while meeting normal output requirements. The reserve constraints for hydropower units are defined in Eq. (52), while the reserve constraints for PHS units under generating and pumping conditions are specified in Eqs. (53) and (54), respectively.

$$\begin{cases} R^{HP(ID)}_{up,k,i,g,t}+R^{HP(ID)}_{PFR,k,i,g,t}+R^{HP(ID)}_{SFR,k,i,g,t}\le u^{HP(ID)}_{k,i,g,t}\cdot P^{HP}_{\max,i,g}-P^{HP}_{k,i,g,t} \\ R^{HP(ID)}_{dn,k,i,g,t}\le P^{HP}_{k,i,g,t}-u^{HP(ID)}_{k,i,g,t}\cdot P^{HP}_{\min,i,g} \end{cases} \tag{51}$$

$$\begin{cases} R^{PS.G(ID)}_{up,k,g,t}+R^{PS.G(ID)}_{PFR,k,g,t}+R^{PS.G(ID)}_{SFR,k,g,t}\le u^{PS.G(ID)}_{k,g,t}\cdot P^{PS.G}_{\max,g}-P^{PS.G}_{k,g,t} \\ R^{PS.G(ID)}_{dn,k,g,t}\le P^{PS.G}_{k,i,g,t}-u^{PS.G(ID)}_{k,g,t}\cdot P^{PS.G}_{\min,g} \end{cases} \tag{52}$$

$$\begin{cases} R^{PS.P(ID)}_{up,k,g,t}+R^{PS.P(ID)}_{PFR,k,g,t}+R^{PS.P(ID)}_{SFR,k,g,t}\le u^{PS.P(ID)}_{k,g,t}\cdot P^{PS.P}_{\max,g}-P^{PS.P}_{k,g,t} \\ R^{PS.P(ID)}_{dn,k,g,t}\le P^{PS.P}_{k,g,t}-u^{PS.P(ID)}_{k,g,t}\cdot P^{PS.P}_{\min,g} \end{cases} \tag{53}$$

(3) Wind and PV output interval constraints. Wind power output modeling is as follows:

$$\begin{cases} 0 \le P_{k,t}^{WT(ID)} \le P_r^{WT} \\ P_{k,\min,t}^{WT(ID)} \le P_{k,t}^{WT(ID)} \le P_{k,\max,t}^{WT(ID)} \end{cases} \tag{54}$$

$$\begin{cases} P_{k,up,t}^{WT(ID)} = P_{f,k,t}^{WT(ID)}(t) + z_{\alpha/2}\sigma_{WT} \\ P_{k,dn,t}^{WT(ID)} = P_{f,k,t}^{WT(ID)}(t) - z_{\alpha/2}\sigma_{WT} \\ P_{k,\max,t}^{WT(ID)} = \min\left\{P_{k,up,t}^{WT(ID)}, P_r^{WT}\right\} \\ P_{k,\min,t}^{WT(ID)} = \max\left\{0, P_{k,dn,t}^{WT(ID)}\right\} \end{cases} \tag{55}$$

where $P_{k,min,t}^{WT(ID)}$ and $P_{k,max,t}^{WT(ID)}$ denote the lower and upper limits of the available power for the wind farm at time t under scenario k, while $P_{k,up,t}^{WT(ID)}$ and $P_{k,dn,t}^{WT(ID)}$ represent the upper and lower limits of the predicted output for the wind farm. The modeling of PV output aligns with the aforementioned wind output and will not be elaborated further.

(4) Multi-level flexibility balancing constraints. These constraints include steady-state power regulation constraint (13), transient frequency security constraints (25–27), and quasi-steady-state frequency recovery flexibility balance constraint (28).

2.5. Solution method

Given the nonlinearity, high-dimensional discontinuity, and multi-objective coupling characteristics of the aforementioned scheduling model, solving it proves to be challenging. To enhance the model's solvability and ensure computational efficiency, this section linearizes the complex constraints and simplifies the original model into a single-objective mixed-integer linear programming (MILP) problem through appropriate mathematical transformations.

2.5.1. Transformation of multi-objective optimization problems

Due to substantial disparities in the magnitude and dimensions of the objective functions within the day-ahead scheduling model, directly addressing the multi-objective optimization problem may significantly elevate computational complexity and impede the identification of feasible solutions that meet all objectives simultaneously. Thus, this study utilizes a weighted minimum norm ideal point method to normalize each sub-objective function's dimensions and assign appropriate weights reflecting their significance. The unified composite objective function is formulated as:

$$\max F = \sum_{i=1}^{3} \omega_i \cdot \left| \frac{\Pi_i - \Pi_i^*}{\Pi_1^{\max} - \Pi_1^*} \right|^2 \tag{56}$$

where $\Pi*1$, $\Pi*2$, and $\Pi*3$are the ideal point target values, representing the expected values of each objective. The coefficients ω_1, ω_2, and ω_3 correspond to the weights of the sub-objective functions Π_1, Π_2, and Π_3, reflecting the significance of each objective in the composite optimization, subject to the constraint $\omega_1 + \omega_2 + \omega_3 = 1$.

2.5.2. Linearization of maximum frequency deviation constraint

The left side of the *MDF* constraint (26) demonstrates significant nonlinearity, influenced by the equivalent frequency response parameters H_{sys}, R_T, F_H, and T_R. The variation in the equivalent reheating time constant T_R exerts a minor effect on frequency fluctuations [42]. It is generally assumed that all units share the same time constant, allowing for simplification of the nonlinear component to a function dependent on three equivalent parameters, as outlined in Eq. (57).

$$G_{sys}(H_{sys}, F_H, R_T) = \frac{D + R_T}{\left(1 + \sqrt{T_R(R_T - F_H)/2H_{sys}}\, e^{-\xi\omega_n t_{nadir}}\right)} \tag{57}$$

$G_{sys}(H_{sys}, F_H, R_T)$ is a nonlinear function of three variables that exhibits local convexity. To address this, a Piecewise linearization (PWL) method is employed to convert the function into a linear form [43]. This method seeks to minimize the objective function (58).

$$\min_{\psi} \sum_{\eta} \left(\max_{1 \le \nu \le \bar{\nu}} \left\{ a_\nu H_{sys}^{(\eta)} + b_\nu F_H^{(\eta)} + c_\nu R_T^{(\eta)} + d_\nu \right\} - G\left(H_{sys}^{(\eta)}, F_H^{(\eta)}, R_T^{(\eta)}\right) \right)^2 \tag{58}$$

where $\Psi = \{a_\nu, b_\nu, c_\nu, d_\nu\}$ denotes the set of optimization variables, ν indicates the number of PWL segments, and η represents the evaluation point. The goal of this optimization problem is to approximate $G_{sys}(H_{sys}, F_H, R_T)$ using a PWL function at each breakpoint. To tackle this min-max problem, new binary variables are introduced to eliminate the inner max operator from the objective function, as defined in Eq. (59), $\bar{\nu}$ where the maximum number of PWL segments is specified.

$$\begin{cases} \gamma_1 = \max\left[a_1 H_{sys}^{(\eta)} + b_1 F_H^{(\eta)} + c_1 R_T^{(\eta)} + d_1, a_2 H_{sys}^{(\eta)} + b_2 F_H^{(\eta)} + c_2 R_T^{(\eta)} + d_2\right] \\ \gamma_{\nu-1} = \max\left[\gamma_{\nu-2}, a_\nu H_{sys}^{(\eta)} + b_\nu F_H^{(\eta)} + c_\nu R_T^{(\eta)} + d_\nu\right], \qquad 3 \le \nu \le \bar{\nu} \end{cases} \tag{59}$$

The big-M method is subsequently applied to transform the unconstrained optimization problem into a constrained mixed-integer quadratic programming problem, as illustrated in Eqs. (60)-(65).

$$\min_{\psi} \sum_{\eta} \left(\gamma\left(H_{sys}^{(\eta)}, F_H^{(\eta)}, R_T^{(\eta)}\right) - G\left(H_{sys}^{(\eta)}, F_H^{(\eta)}, R_T^{(\eta)}\right) \right)^2 \tag{60}$$

$$\begin{cases} a_1 H_{sys}^{(\eta)} + b_1 F_H^{(\eta)} + c_1 R_T^{(\eta)} + d_1 \le \gamma_1 \\ \gamma_1 \le a_1 H_{sys}^{(\eta)} + b_1 F_H^{(\eta)} + c_1 R_T^{(\eta)} + d_1 + \delta_1 M \end{cases} \tag{61}$$

$$\begin{cases} a_2 H_{sys}^{(\eta)} + b_2 F_H^{(\eta)} + c_2 R_T^{(\eta)} + d_2 \le \gamma_1 \\ \gamma_1 \le a_2 H_{sys}^{(\eta)} + b_2 F_H^{(\eta)} + c_2 R_T^{(\eta)} + d_2 + (1-\delta_1) M \end{cases} \tag{62}$$

$$\gamma_{\nu-2} \le \gamma_{\nu-1} \le \gamma_{\nu-2} + \delta_{\nu-1} M \tag{63}$$

$$\begin{cases} a_\nu H_{sys}^{(\eta)} + b_\nu F_H^{(\eta)} + c_\nu R_T^{(\eta)} + d_\nu \le \gamma_{\nu-1} \\ \gamma_{\nu-1} \le a_\nu R_\nu^{\eta} + b_\nu F_\nu^{\eta} + c_\nu H^{\eta} + d_\nu + (1-\delta_{\nu-1}) M \end{cases} \tag{64}$$

$$\underline{a} \le a_\nu \le \bar{a}, \underline{b} \le b_\nu \le \bar{b}, \underline{c} \le c_\nu \le \bar{c}, \underline{d} \le d_\nu \le \bar{d} \tag{65}$$

where δ is binary variable, M is a sufficiently large positive number.

The PWL accuracy can be regulated by adjusting the number of evaluation points, segments, and the feasible domain of the four coefficients (Eq. (65)). Fitting results for $\nu = 4$ and $R_T = 20$, shown in Fig. 9, demonstrate that the PWL method limits the maximum

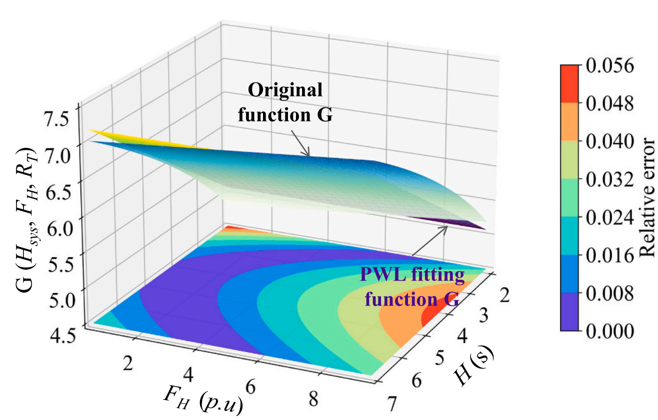


Fig. 9. The upper figure depicts the function G_{sys} (H_{sys}, F_H, R_T) for $R_T = 20$, while the lower figure illustrates the relative error of the approximation.

linearization error to below 6 %.

Upon obtaining the optimal solution ($a^* v$, $b^* v$, $c^* v$, $d^* v$) for the linearized model of the function G_{sys}, constraint (26) can be transformed into $4^*(\overline{v}-1)$ linear inequality constraints that ensure the maximization of the value $\gamma_{\overline{v}-1}$, along with the $\gamma_{\overline{v}-1,t}$ threshold constraint

$$R^{sys}_{PFR,t}+\Delta f^{\max}_{nadir}\cdot\gamma_{\overline{v}-1,t}/f_0 \geq |\Delta P_{d,t}|. \tag{66}$$

2.5.3. Model solution process

Following the aforementioned transformations, the day-ahead delivery curve optimization and intraday scheduling models are formulated as MILP problems, suitable for resolution using Gurobi solvers. A hierarchical scheduling approach is employed to optimize decisions across multiple time scales, effectively balancing system economy and frequency stability. Specifically, the day-ahead optimization model utilizes typical forecast-based scenarios to maximize system benefits while ensuring output stability, thereby establishing dynamic boundaries for the 24-h power delivery of the integrated system. Subsequently, the simulation resolution is refined to 15-min intervals. Upon issuing the power delivery plan, intraday simulations are conducted, with decision variables including the start-up, peak regulation, and frequency reserve capacities of cascade hydropower and PHS units, ultimately yielding the optimal output process across 96-time intervals.

3. Optimization of VRE integration capacity in the CHIDS

The integration capacity of VRE is defined as the CHIDS's ability to incorporate wind and PV power under specific operational conditions, typically quantified by the VRE penetration level (η_{VRE}), as shown in Eq. (67). As η_{VRE} increases, the power system faces greater uncertainty and reduced controllability, challenging system flexibility. This section presents a framework for assessing system flexibility improvements, aiming to optimize VRE integration capacity across varying η_{VRE} levels, balancing efficiency and reliability.

$$\eta_{VRE}=\frac{Cap_{VRE}}{Cap_{sys}}=\frac{P^{WT}_r+P^{PV}_r}{Cap_{sys}}\times 100\%. \tag{67}$$

3.1. Assessing system flexibility enhancement: a multidimensional framework

Table 1 provides the definitions and calculation methods for key indicators analyzed across three dimensions: frequency response security, power delivery reliability, and operational economy. These indicators assess the system's effectiveness in managing increased VRE integration.

3.2. Optimization method for VRE integration capacity

The optimization method for the VRE integration capacity of the CHIDS, based on quantifiable indicators of system flexibility enhancement, is shown in Fig. 10. This method includes the following key modules: model parameters input, wind and PV output forecasting, multi-timescale flexibility optimization scheduling, linear approximation of frequency security constraints, multidimensional assessment of system flexibility enhancement, and optimal VRE penetration level decision-making. The specific steps are as follows:

Step 1: Input the CHIDS's fundamental parameters, including installed capacities of renewable energy sources, historical generation data, receiving-end load demand, and other variables.
Step 2: Define the simulation interval for VRE penetration rates [ηmin VRE, ηmax VRE] and the simulation step size $\Delta\eta_{VRE}$, initializing the total installed capacity of VRE $Cap_{VRE,0}$ when $\eta_{VRE}=\eta$min VRE.
Step 3: Utilize meteorological data and historical generation records to create a typical scenario database for wind and PV day-ahead output, and perform intra-day confidence interval predictions.
Step 4: Optimize the multi-time-scale scheduling strategy of the CHIDS to enhance economic performance and flexibility by considering wind and PV output fluctuations.
Step 5: Calculate the frequency security indicators for all typical scenarios based on simulation results and evaluate their compliance with specified threshold requirements. If they meet the criteria, proceed to Step 6; otherwise, adjust the feasible region of the

Table 1
Key indicators and calculation methods for evaluating system flexibility.

Dimension	Secondary indicators	Definition	Calculation method
Frequency response security	*RoCoF* violation gap (VD_{RoCoF})	It is employed to assess the *RoCoF* exceeds the threshold in the early stage of the disturbance, reflecting the system's inertia and rapid response capability.	$VD_{RoCoF}=(RocoF_t-RocoF_{\max})\times 100\%$
	MFD violation gap ($VD_{\Delta f\text{nadir}}$)	It refers the extent to which frequency deviations exceed safety thresholds, assessing the system's dynamic stability and regulation capability.	$VD_{\Delta f\text{nadir}}=\left(\Delta f_{\text{nadir},t}-\Delta f^{\max}_{\text{nadir}}\right)\times 100\%$
	QSFD recovery probability (RP_{QSFD})	It refers to the probability that the system's frequency deviation returns to a safe range after a disturbance, used to assess the system's quasi-steady-state frequency recovery capability.	$RP_{QSFD}=\sum_{t\in T}\delta_{safe}(t)/T$ If $\lvert\Delta f_{ss}(t)\rvert\leq\Delta f^{\max}_{ss},\delta_{safe}(t)=1;\text{else},\delta_{safe}(t)=0$
Power delivery reliability	Flexibility insufficiency probability (*FIP*)	It represents the proportion of time when the system cannot meet regulation demand due to insufficient flexibility resources, divided into *UFIP* and *DFIP*.	$FIP=UFIP+DFIP$ $UFIP=\sum_{t\in T}\delta_{UF}(t)/T, DFIP=\sum_{t\in T}\delta_{DF}(t)/T$
	Renewable energy curtailment rate (*ECR*)	It refers to the proportion of wind and PV generation that is curtailed due to insufficient regulation capacity or transmission constraints.	$ECR=\frac{\sum_{t\in T}P^{cur}(t)}{\sum_{t\in T}\left(P^{WT}_{f,t}+P^{PV}_{f,t}\right)}\times 100\%$
	Power shortage rate (*PSR*)	It refers to the proportion of unmet power delivery demand due to insufficient regulation capacity, relative to the total demand.	$PSR=\frac{\sum_{t\in T}P^{loss}(t)}{\sum_{t\in T}P^{DC}(t)}$
	Annual effective utilization hours of PHS (EUR_{PHS})	It refers to the ratio of the actual annual generation of a PHS station to its installed capacity, reflecting the efficiency and frequency of the station's utilization of its generation capacity.	$EUR_{PHS}=\frac{\sum_{t\in T}\sum_{g\in N_{PS}}P^{PS.G}_g(t)}{Cap_{PS}}$
Operational economy	Levelized cost of electricity (*LCOE*)	It is a key indicator of a power generation project's economic viability, representing the average cost per kWh needed to achieve a net present value (NPV) of zero over its lifetime [10].	$LCOE=\frac{C_{inv}\frac{(1+r)^Y-1}{r(1+r)^Y}+\sum_{y=1}^{Y}\frac{C_{O\&M,y}}{(1+r)^y}-C_{sal}\frac{(1+r)^Y-1}{r(1+r)^Y}}{\sum_{y\in Y}\sum_{t\in T}\sum_{g\in Nu}\frac{P_{i,t}}{(1+r)^y}\cdot\Delta t}$
	Internal rate of return (*IRR*)	It is the discount rate that makes the project's NPV zero, reflecting the rate of return and key for evaluating profitability and feasibility.	$NPV=\sum_{y=0}^{Y}(CI-CO)_y(1+IRR)^{-y}=0$

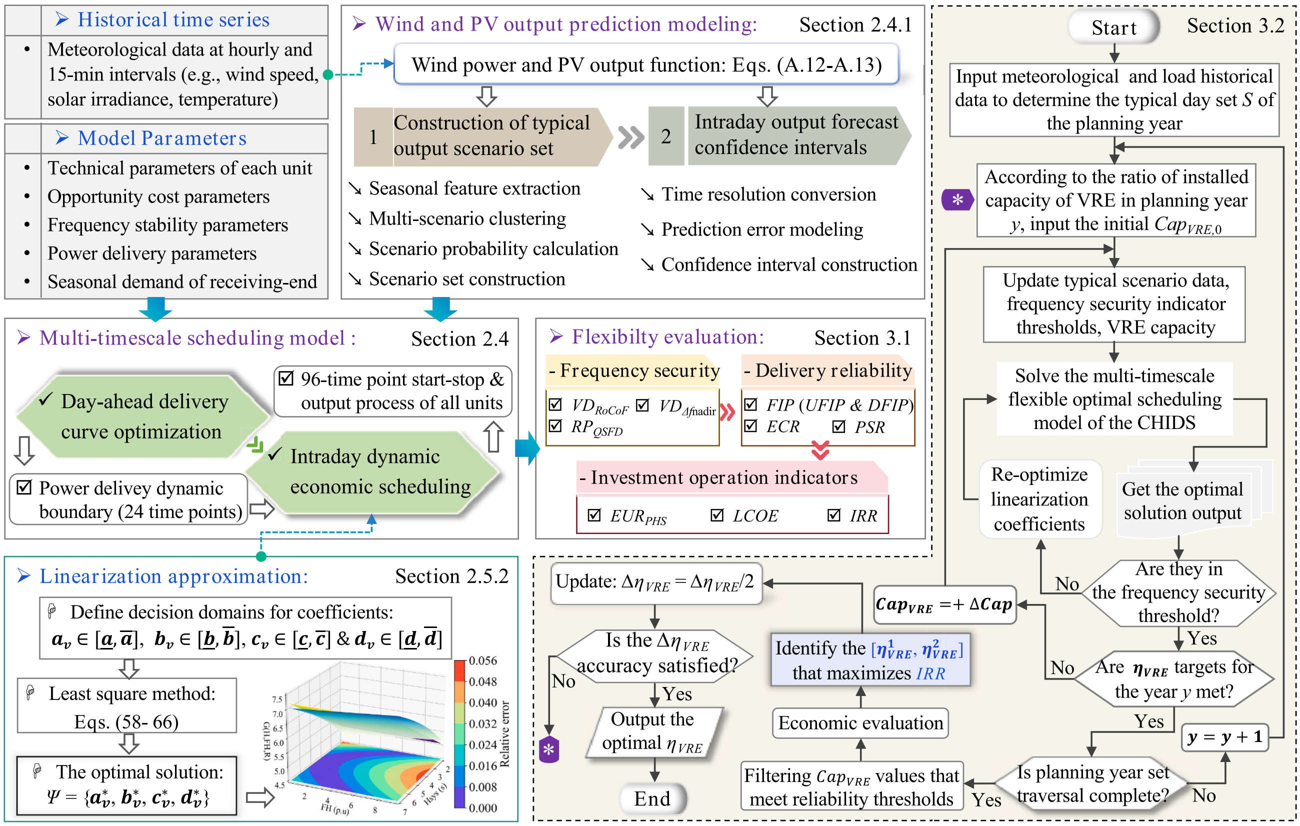


Fig. 10. Overall framework for identifying the integration capacity of variable renewable energy in the cascade hydro-wind-solar-storage integrated delivery system.

linearization coefficients $\{a_v, b_v, c_v, d_v\}$ and return to Step 4 for further optimization.

Step 6: Utilize annual simulation results across various η_{VRE} levels to estimate power delivery reliability indicators. Concurrently, filter penetration rate sets that meet established reliability criteria and use operational economic indicators for investment estimation, informing decision-making.

Step 7: Identify the range of penetration rates [η1 VRE, η2 VRE] that maximizes the *IRR* for the CHIDS. Update the step size $\Delta\eta_{VRE} = \Delta\eta_{VRE}/2$ and assess whether the η_{VRE} meets precision requirements. If so, identify the corresponding η_{VRE} value that yields the maximum *IRR* as the optimal VRE integration capacity and output the results. If not, set ηmin VRE = η1 VRE and ηmax VRE = η2 VRE, then return to Step 2.

4. Case study

This model is applied to a renewable energy hub in Qinghai Province, located in the upper Yellow River basin, an area abundant in natural resources, particularly wind, solar, and hydropower, offering significant development potential. The region's strategic position facilitates the creation of an integrated hydro-wind-solar-storage delivery system. This system includes several cascade hydropower stations, a planned PHS facility, and nearby wind and PV stations. Fig. 11 illustrates the geographical distribution and network topology of the power stations within the system. Clean energy is efficiently transmitted to provinces, including Henan, via the Qingyu UHVDC transmission project, which has a transmission capacity of 8000 MW and operates at a voltage of ±800 kV.

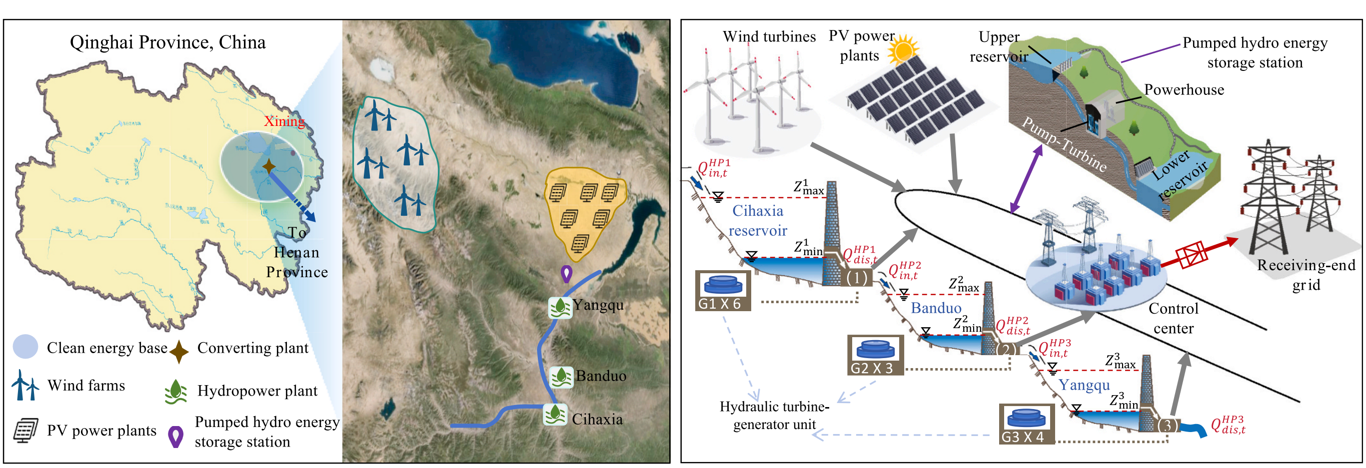


Fig. 11. Geographic distribution and network topology of the clean energy hub.

4.1. Model parameters

(1) Cascade hydropower stations: Three cascade reservoirs—Cihaxia, Bando, and Yangqu—have been constructed in the upper Yellow River basin. These reservoirs differ significantly in water storage capacity and regulation capabilities, supporting seasonal, daily, and weekly adjustments [44]. The operational parameters for each power station, as presented in Table D1, include reservoir characteristics, hydropower unit performance metrics, frequency response model parameters, and operational cost parameters [45].

(2) Wind and PV power stations: The site is currently undergoing scale optimization, with a target increase to 18 GW. Due to limited historical data, wind and solar simulation data are derived from meteorological information published by the Global Wind Atlas and PVGIS and transformed using their output models (Eqs. (A.12)-(A.13)). Historical planning documents suggest optimal complementarity occurs when the ratio of wind power and PV capacity in Qinghai Province is between 1.5 and 2.0. Thus, the preliminary installed capacities are set at 4700 MW for wind and 8000 MW for PV.

(3) Pumped hydro storage power station: The proposed upper reservoir is designed to maintain a normal water level of 3645 m and a regulation capacity of 13.65 million m^3. The lower reservoir, with a normal water level of 3255 m, possesses the same regulation capacity, facilitating daily regulation. The station's total planned installed capacity is 2700 MW, comprising nine reversible PHS units, each rated at 300 MW. The specifications for FS-PHS and VS-PHS units are based on reference [46], the frequency response model parameters are derived from reference [47], and the operational cost parameters are sourced from references [48,49], as summarized in Table 2. Investment costs for VS-PHS are expected to increase by 18 % compared to FS-PHS [9].

(4) Frequency security parameters: According to the national standard "GB/T 40595–2021" [50], the rated frequency of the power grid is 50 Hz, with a permissible steady-state frequency deviation limit of ±0.2 Hz. The *RoCOF* is capped at 0.5 Hz/s, and the *MFD* threshold is 0.5 Hz.

Table 2
Parameters of the pumped hydro storage power station.

Parameters	Symbol	Unit	FS-PHS	VS-PHS
Number of hydro units	N_{PS}	–	9	9
Maximum generation/pumping power	$P_{max}^{PS.G}$ / $P_{max}^{PS.P}$	MW	300	300
Minimum generation power	$P_{min}^{PS.G}$	MW	120	60
Minimum pumping power	$P_{min}^{PS.P}$	MW	300	150
Ramping speed	ΔP_{ramp}^{PS}	MW/min	5	10
Limit on start-stop cycles	M_{PS}	p.u.	4	6
Minimum operation/ shutdown duration	T_{on}^{PS}/ T_{off}^{PS}	min	30	15
Synthetic cycle efficiency	η^{PS}	%	75	80
Inertial time constant	H_{PS}	s	10	5
Equivalent turbine parameter	F_{H}^{PS}	p.u.	0.25	0.83
Equivalent governor regulation constant	R_{T}^{PS}	p.u.	0.04	0.04
Startup cost	C_{on}^{PS}	CNY	30	40
Peak-shaving reserve cost	C_{peak}^{PS}	CNY/MWh	2.8	3
Primary frequency regulation reserve cost	C_{PFR}^{PS}	CNY/MWh	3.2	3.7
Secondary frequency regulation reserve cost	C_{SFR}^{PS}	CNY/MWh	2.8	3

4.2. Uncertainty scenarios for wind and solar output

This section utilizes the uncertainty modeling methods for wind and solar output described in Section 2.4.1. Annual data of normalized wind and solar power from a typical year, depicted in Fig. 12(a), is categorized by season and clustered into 24 sub-scenarios, as shown in Fig. 12(b), to support day-ahead optimization simulations. Table 3 presents the probabilities of six typical scenarios within each season. Using the Bootstrap confidence interval method, intraday output forecasts for wind and solar power at 15-min intervals are generated for simulation analysis. Fig. 12(c) illustrates the confidence intervals for wind and solar output predictions on a typical spring day.

4.3. Simulation scenarios setting

To validate the proposed multi-timescale optimization model's effectiveness, three distinct scheduling models were developed for comparative analysis. While these models share the same day-ahead optimization model, they differ in the intra-day optimization phase, as described below:

(1) Traditional constraint unit commitment (TCUC) model: This model employs steady-state power regulation as the sole flexibility constraint during the intra-day optimization phase, omitting frequency regulation reserve costs from the objective function.

(2) Security-constrained unit commitment (SCUC) model: In this model, both power regulation and frequency response are incorporated as flexibility constraints for intra-day optimization. However, frequency regulation reserve costs are excluded. The reserve capacity for frequency control is determined based on the droop coefficients of each unit, as specified in Eq. (68) [37].

$$P_{g,\max} - P_{g,t} \geq \frac{u_{g,t}P_{g,\max}/R_{T,g}}{\sum_{g \in Nu} u_{g,t}P_{g,\max}/R_{T,g}}\Delta P_{d,t} \tag{68}$$

where *Nu* is the total number of units within CHIDS that possess frequency regulation capability.

(3) Flexibility-balanced unit commitment (FBUC) model: This model adopts the dynamic economic optimization scheduling strategy proposed in this study. It integrates steady-state power regulation, transient frequency response, and quasi-steady-state frequency recovery as flexibility constraints, while incorporating frequency regulation reserve costs into the objective function.

Table 4 delineates the objective functions and constraints associated with the intra-day optimization scheduling models outlined above. Additionally, two simulation cases for the CHIDS are constructed, one incorporating FS-PHS units and the other incorporating VS-PHS units. A comparative analysis is then conducted to assess the impact of these configurations on the operational performance of the integrated delivery system.

5. Results and discussion

5.1. Analysis of system optimal scheduling results

This section examines the optimal scheduling results of the CHIDS for a representative spring day, utilizing two simulations: one with FS-PHS and the other with VS-PHS units. The purpose of the analysis is to validate the proposed model's effectiveness and assess how PHS flexibility influences the operational efficiency and stability of the CHIDS.

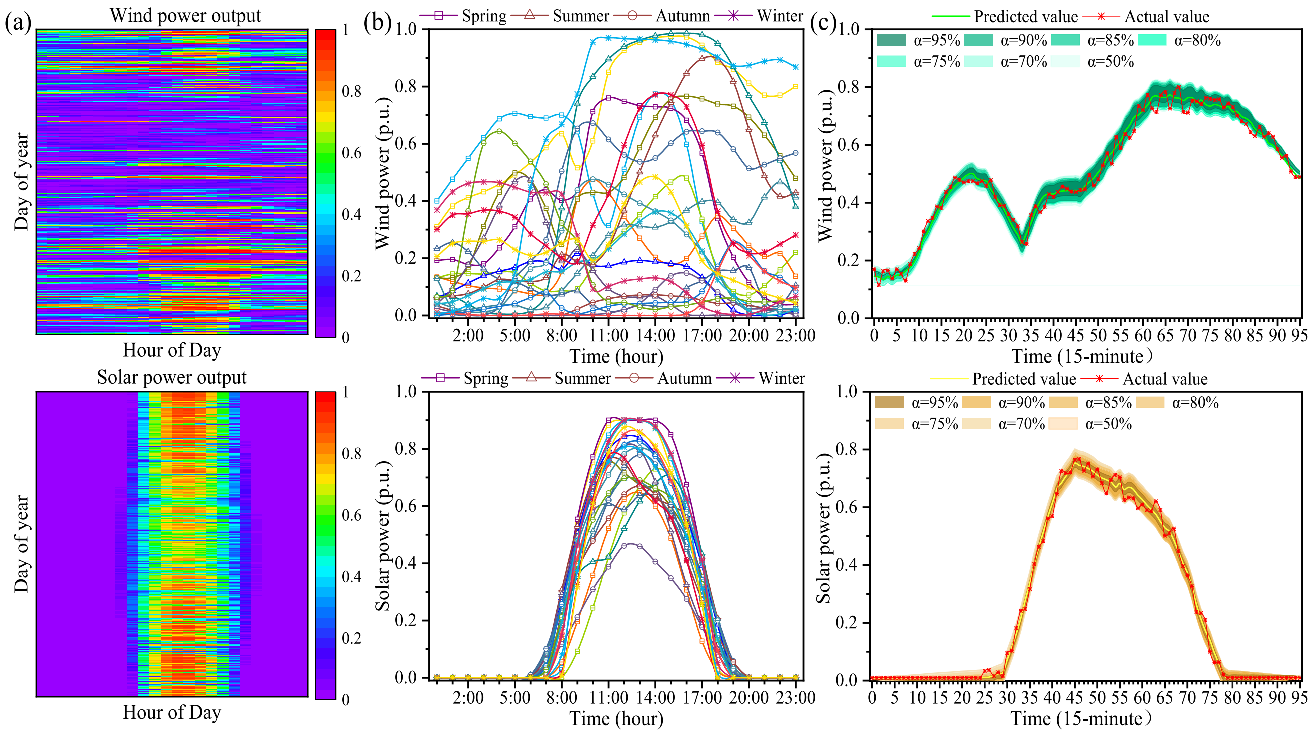


Fig. 12. Analysis of wind and solar power output: (a) Annual normalized output; (b) Seasonal typical output scenarios; and (c) Confidence intervals for intra-day output forecasts.

Table 3
Seasonal distribution of wind and solar power scenarios.

Season	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6
Spring (Mar-May)	0.08	0.19	0.28	0.27	0.08	0.11
Summer (Jun-Aug)	0.20	0.11	0.11	0.16	0.22	0.21
Autumn (Sep-Nov)	0.12	0.11	0.26	0.20	0.28	0.03
Winter (Dec-Feb)	0.22	0.09	0.31	0.09	0.08	0.22

Table 4
Objective functions and constraints of the different intraday optimization models.

Model	Objective Function	Shared Constraints	Unique Constraints
TCUC	$\min F_{sys} = \sum_{k\in N_k}\sum_{t\in N_T^{ID}} p_k\cdot\begin{pmatrix} F_{k,t}^{on}+F_{k,t}^{off}+ \\ F_{k,t}^{PRC}+F_{k,t}^{Risk}\end{pmatrix}\cdot\Delta t^{(\mathrm{ID})}$		None
SCUC	$\min F_{sys} = \sum_{k\in N_k}\sum_{t\in N_T^{ID}} p_k\cdot\begin{pmatrix} F_{k,t}^{on}+F_{k,t}^{off}+ \\ F_{k,t}^{PRC}+F_{k,t}^{Risk}\end{pmatrix}\cdot\Delta t^{(\mathrm{ID})}$	Eqs. (A.1)-(A.11), Eqs. (A.14)-(A.24), Eq. (13), and Eqs. (51)–(55)	Eq. (25), Eqs. (16)–(17), and Eq. (68)
FBUC	$\min F_{sys} = \sum_{k\in N_k}\sum_{t\in N_T^{ID}} p_k\cdot\begin{pmatrix} F_{k,t}^{on}+F_{k,t}^{off}+F_{k,t}^{PRC} \\ +F_{k,t}^{FRC}+F_{k,t}^{Risk}\end{pmatrix}\cdot\Delta t^{(\mathrm{ID})}$		Eqs. (25)–(28)

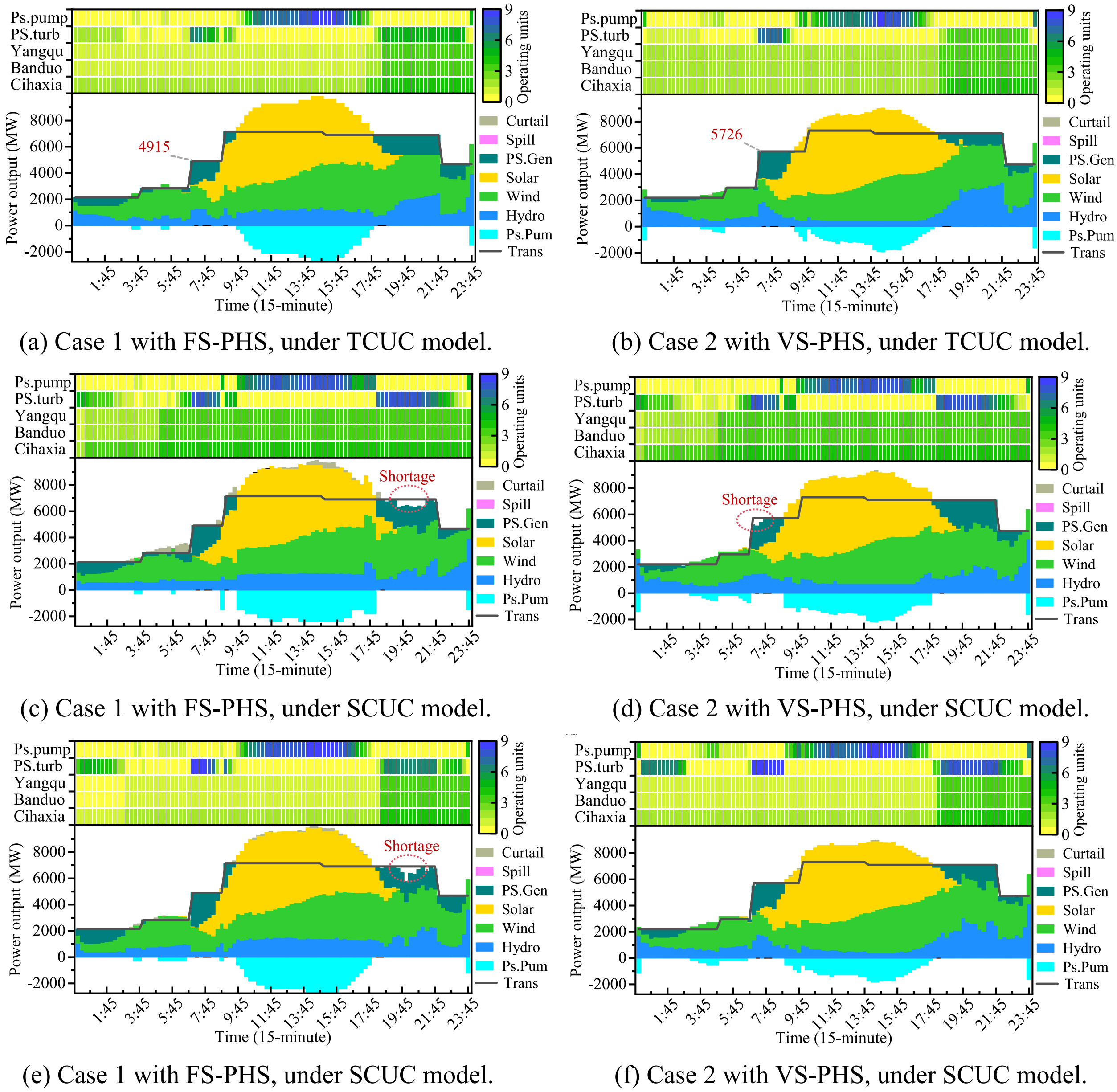
5.1.1. *Optimal scheduling results under various operational scenarios*

Fig. 13 illustrates the operational characteristics of two types of PHS unit configurations under three dispatch models during a typical spring day. The analysis covers 15-min interval delivery power tracking, multi-energy generation coordination, and the dynamic startup/shutdown behavior of hydro and PHS units.

The results indicate that VS-PHS units in Case 2, with their broader power regulation range and continuous speed control capabilities, significantly enhance system flexibility. During mid-to-high power delivery periods (07:14 to 08:45), Case 2's day-ahead transmission plan achieves a 16.5 % improvement over Case 1, highlights the advantages of VS-PHS units in enhancing the economic viability of power transmission.

A comparative analysis of the three dispatch models presented in Fig. 13 reveals that the TCUC model exhibits pronounced security deficiencies in its quest for economic optimization. The absence of frequency security constraints in the TCUC model results in the lowest curtailment and power shortage rates among the three models, confirming that minimizing unit startup costs and reserve configuration costs achieves optimal operational economy. However, this exclusive focus on economic efficiency compromises the system's inertia support, thereby exposing it to potential frequency instability during disturbances.

In contrast, the SCUC model significantly enhances grid dynamic security by incorporating transient frequency response constraints. This improvement, however, comes at the expense of system flexibility and



(a) Case 1 with FS-PHS, under TCUC model.
(b) Case 2 with VS-PHS, under TCUC model.
(c) Case 1 with FS-PHS, under SCUC model.
(d) Case 2 with VS-PHS, under SCUC model.
(e) Case 1 with FS-PHS, under SCUC model.
(f) Case 2 with VS-PHS, under SCUC model.

Fig. 13. Power transmission plans, energy generation portfolios, and unit operational states for the CHIDS during the spring for two case scenarios under three different unit operational models. The upper-panel heatmap illustrates online unit numbers, while the lower-panel stacked chart depicts energy profiles, with negative values indicating storage charging of PHS.

economic performance. For instance, in Case 1, to satisfy the minimum inertia requirement, the model mandates continuous operation of cascaded hydropower units from 05:00 to 24:00, thereby reducing the adjustment margin available to the PHS units. This rigid constraint yields two notable consequences: first, during periods when transmission capacity is limited, the system must strategically curtail a portion of delivery dispatch power to satisfy frequency regulation reserve requirements, resulting in a curtailment rate of 2.13 %; second, during the evening peak period, a sustained power shortfall lasting 3.25 h is observed, with a maximum shortage of 1492 MW. Conversely, in Case 2, the broad adjustment capabilities of variable-speed units enable rapid pumping to absorb surplus power, reducing the curtailment rate to zero and shifting the power shortfall to the moderate demand interval of 06:45–07:30, with the maximum shortage decreasing to 662 MW. These findings reveal the inherent conservatism of the SCUC model—where excessive allocation of adjustment margin for ensuring frequency security reflects the limitations of conventional security constraints in balancing operational stability and economic efficiency.

The FBUC model addresses these limitations by introducing a three-dimensional constraint framework to achieve synergistic optimization of power system security and economic efficiency. In contrast to the conventional SCUC model, the FBUC framework demonstrates substantial performance enhancements. At the unit dispatch level, the frequency of start-up events for both hydropower and PHS units is significantly reduced. Operationally, Case 1 exhibits a reduced renewable curtailment rate of 0.54 % and shortens power shortage durations to 1.5 h, while Case 2 achieves zero power shortage operation with a near-zero curtailment rate. These results demonstrate that the FBUC model more effectively leverages system flexibility resources to maintain frequency stability, thereby minimizing operational costs without compromising security.

5.1.2. Frequency security analysis under diverse operational scenarios

To evaluate the capability of the scheduling models to ensure the dynamic frequency security of the grid, this study conducts a comparative analysis of frequency security indicators for three models—namely, TCUC, SCUC, and FBUC—across 96 time intervals on a representative spring day, considering two distinct cases. In accordance with the GB/T 40595–2021 standard, the critical safety thresholds are defined as follows: a maximum RoCoF of 0.5 Hz/s, a *MFD* limit of ±0.5 Hz, and a *QSFD* threshold of ±0.2 Hz. As depicted in Fig. 14, the three models exhibit pronounced disparities in their dynamic frequency characteristics.

In the TCUC model, which omits frequency security constraints, both cases manifest severe violations of frequency safety limits. In Case 1, the probability of *MFD* exceeding the threshold reaches 80 %, with the minimum system frequency plunging to 40.02 Hz; additionally, *QSFD* surpasses the ±0.2 Hz limit in 41 intervals, with a peak deviation of 0.363 Hz observed at 09:15, corresponding to an elevated *RoCoF* of 0.576 Hz/s. In Case 2, the incidence of *MFD* violations further escalates to 95 %, with the minimum frequency descending to 40.91 Hz—thereby exposing the inertia coupling effects induced by rapid power modulation in variable-speed units. Although the number of *QSFD* exceedance intervals diminishes to 21, persistent breaches occur between 05:00 and 06:45, with deviation ranges spanning 0.244 to 0.307 Hz. This phenomenon is primarily attributed to the TCUC model's excessive emphasis on economic optimization, achieved by reducing the number of online units relative to the SCUC model, which in turn substantially depletes the system's inertia reserve. Consequently, under external disturbances, the system fails to furnish adequate dynamic response capacity, thereby undermining frequency stability and revealing significant operational limitations.

In contrast, both the SCUC and FBUC models successfully confine all frequency security indicators within the prescribed safety thresholds by incorporating explicit frequency security constraints. Specifically, these models limit *RoCoF* to 0.25 Hz/s for the SCUC model and 0.49 Hz/s for the FBUC model, thereby outperforming the national standard of 0.5 Hz/s. Although the FBUC model exhibits a marginally higher *RoCoF* compared to the SCUC model, it remains within the acceptable safety margins; moreover, its performance with respect to *MFD* and *QSFD* is virtually indistinguishable from that of the SCUC model. This outcome is attributable to the FBUC model's integrative approach, which concurrently accounts for power regulation capability, transient frequency response, and quasi-steady-state frequency recovery constraints. As a result, the model maximizes the utilization of the system's flexibility resources, thereby achieving a synergistic optimization between frequency security and economic efficiency.

5.1.3. Analysis of system flexibility regulation resource balance

Under the proposed FBUC model, the regulation resources of CHIDS are collaboratively provided by cascade hydropower and PHS units. Through the dynamic allocation of peak regulation, PFR, and SFR reserve resources, a multi-objective optimization that simultaneously enhances frequency stability and economic performance is achieved. Fig. 15 illustrates the distribution of upward and downward regulation resources on a typical spring day for two scenarios, as well as the occurrence of power shortages and curtailment resulting from insufficient flexibility.

In Case 1, FS-PHS units, due to their constant-power pumping mode, exhibited pronounced spatiotemporal imbalances in flexibility provision. As depicted in Fig. 15(a), a precipitous decline in VRE output between 18:30 and 22:00 led to a surge in upward regulation demand, resulting in an aggregated power shortage lasting three hours with a maximum deficit of 1148.7 MW. During this period, the supply structure for upward regulation was characterized by a highly polarized distribution, with the Cihaxia hydropower station accounting for 75 % of the peak regulation reserve. Concurrently, during the period of high photovoltaic generation (9:45–13:45), as depicted in Fig. 15(b), the system's utilization of downward flexibility resources approached saturation, culminating in a daily curtailment of 559 MWh. The downward regulation resources were primarily provided by a combination of hydropower stations and PHS units operating in generating mode. Regarding frequency regulation reserves, the PHS units in generating mode, in conjunction with the Yangqu and Bando stations, supplied critical frequency regulation support to the system.

In Case 2, the integration of VS-PSH units significantly improved the spatiotemporal distribution of regulation resources. The variable-speed units reduced the upward capacity shortfall on a typical day to 227 MWh (Fig. 15(c)) and decreased curtailment to 26.53 MWh (Fig. 15(d)), representing an approximate 95.3 % reduction compared to Case 1. Upward flexibility is primarily provided by the Cihaxia and Yangqu stations, along with VS-PSH in generation mode, ensuring the reliability and stability of delivery power dispatch during periods of low VRE output. Conversely, downward flexibility, with 71 % supplied by VS-

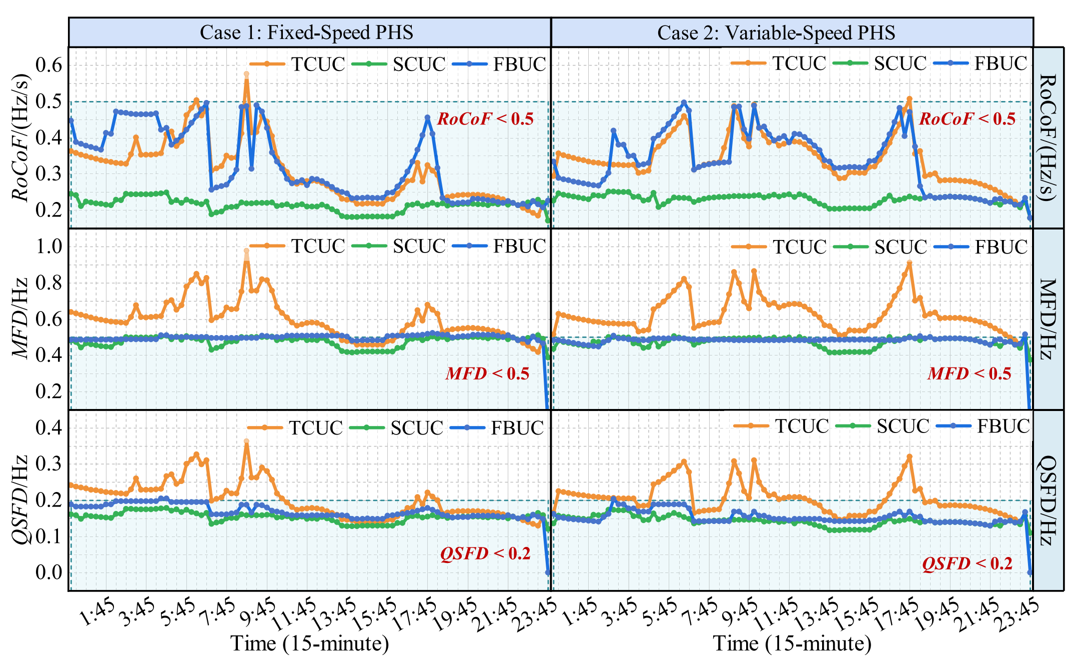


Fig. 14. Comparison of frequency security indicators across different operational models for two cases on a typical spring day.

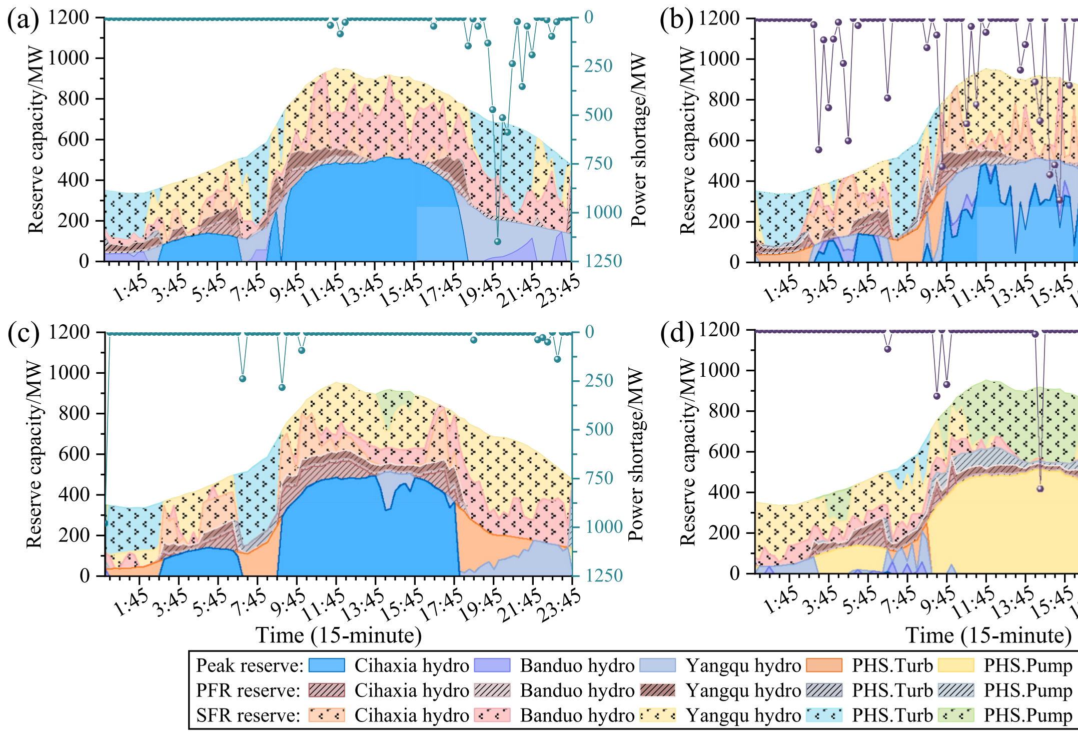


Fig. 15. Distribution of system upward and downward flexibility regulation resources under the FBUC model on a typical spring day for two scenarios. (a) Upward regulation resource distribution in Case 1, (b) downward regulation resource distribution in Case 1, (c) upward regulation resource distribution in Case 2, and (d) downward regulation resource distribution in Case 2.

PSH in pump mode, enhances VRE absorption during peak delivery periods. Additionally, downward frequency reserve is supported by the Yangqu and the VS-PSH in both generation and pump modes, ensuring grid frequency stability and operational reliability.

5.2. Evaluation of the flexibility contribution of PHS under different operating scenarios

This section uses the optimized results of CHIDS without PHS in the TCUC operation mode as a baseline. It quantifies the flexibility contributions of both FS-PHS and VS-PHS across the TCUC, SCUC, and FBUC

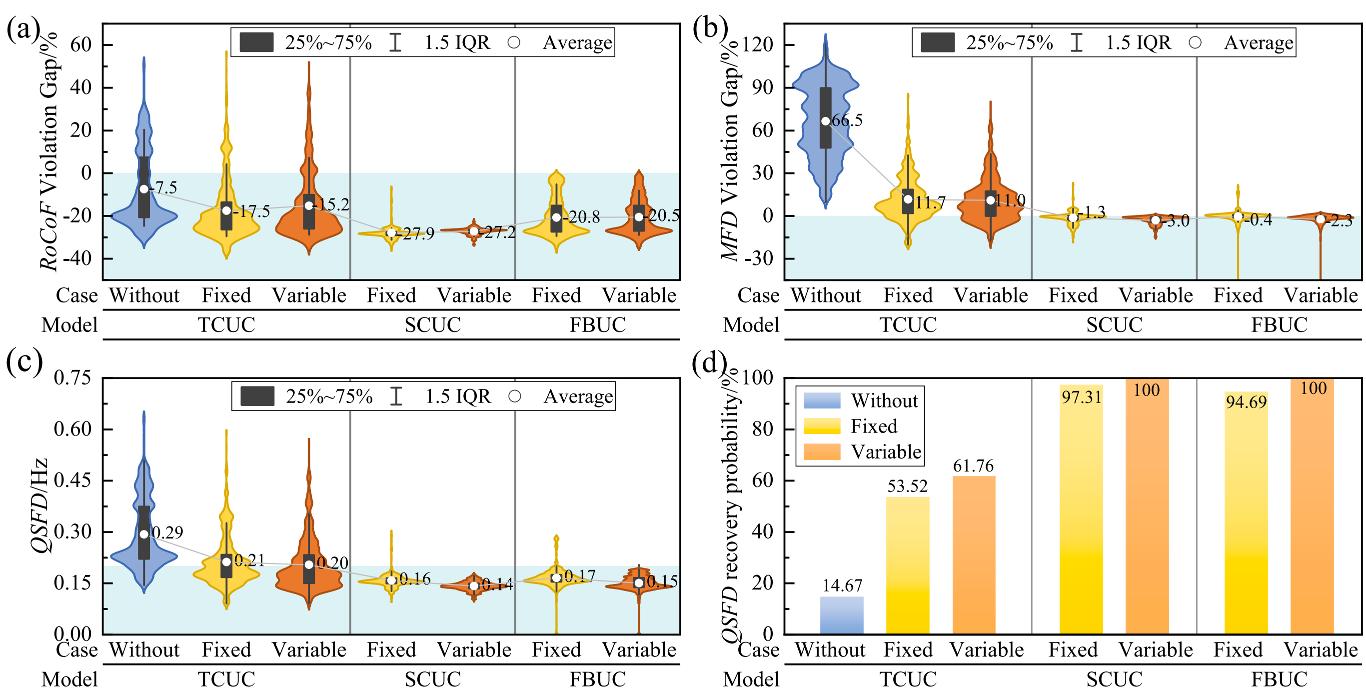


Fig. 16. Distribution of frequency security assessment indicators for various cases under three operational modes. (a) *RoCoF* violation gap, (b) *MFD* violation gap, (c) *QSFD*, and (d) Probability of *QSFD* recovery.

operation modes, evaluating them from the perspectives of frequency security, delivery reliability, and operational economy. Through comparative analysis, the influence of different operational modes on PHS regulation capabilities and its synergy within the system is elucidated. Notably, when the *MFD* threshold is set at 0.5 Hz, no feasible solutions exist for the SCUC and FBUC models, indicating the system's inability to meet safety and reliability requirements under high flexibility constraints. Consequently, results from SCUC and FBUC modes are excluded from further evaluations to maintain scientific consistency and logical integrity.

5.2.1. Frequency security assessment

To quantitatively assess the contribution of FS-PHS and VS-PHS to frequency security, this study analyzes the distribution of four frequency security indicators over a one-year time scale under three different operating modes: *RoCoF* violation gap (VD_{RoCoF}), *MFD* violation gap ($VD_{\Delta f\mathrm{nadir}}$), *QSFD*, and the probability of *QSFD* recovery (RP_{QSFD}), as shown in Fig. 16.

The distribution of VD_{RoCoF} in Fig. 16(a) indicates the system's frequency response performance across different scenarios. In the TCUC mode, without the involvement of PHS, the interquartile range (IQR) of the VD_{RoCoF} values spans from −20 % to 8 %. This suggests that, although the system's *RoCoF* remains below the threshold in most cases, some periods still exceed safety limits, highlighting insufficient frequency response capability without PHS support. While the inclusion of FS-PHS and VS-PHS improves the VD_{RoCoF} distribution in TCUC by shifting most values below zero, some periods still show positive VD_{RoCoF} values, indicating that PHS participation enhances frequency response but does not fully eliminate instability risks. In contrast, both SCUC and FBUC modes, which incorporate frequency constraints, maintain VD_{RoCoF} values below zero across all cases. This demonstrates that both models effectively constrain frequency and its rate of change within safe limits, significantly improving frequency response. Specifically, in the SCUC mode, the IQR of VD_{RoCoF} values for FS-PHS and VS-PHS cases is centered around −27.9 % and − 27.2 %, respectively, with tighter distributions further from zero. These results reflect the superior stability and reliability of SCUC in controlling the *RoCoF*.

As shown in Fig. 16(b), in the TCUC mode, the $VD_{\Delta f\mathrm{nadir}}$ values, representing deviations from the MFD threshold, are above the zero axis in all cases: without PHS, with FS-PHS, and with VS-PHS. Specifically, in the absence of PHS, the average $VD_{\Delta f\mathrm{nadir}}$ value is 66.5 %, significantly exceeding the zero axis. This indicates that, during disturbances, the system's frequency deviation exceeds safety limits, highlighting a substantial decline in stability without PHS support. Consequently, the absence of PHS increases the risk of frequency instability. In contrast, in both SCUC and FBUC modes, PHS participation improves $VD_{\Delta f\mathrm{nadir}}$, with narrower interquartile ranges and values closer to the average. In the SCUC mode, the average $VD_{\Delta f\mathrm{nadir}}$ values for FS-PHS and VS-PHS are −1.3 % and − 3.0 %, respectively, demonstrating effective frequency control. Similarly, in the FBUC mode, the values are −0.4 % and − 2.3 %, further confirming the effectiveness of the proposed frequency-constrained optimization model.

As shown in Fig. 16(c), in the TCUC operating mode, the *QSFD* values for all three cases exceed the 0.2 Hz safety threshold in most periods. This indicates inadequate frequency recovery capability, as the system fails to return swiftly to a stable state following disturbances. Specifically, from Fig. 16(d) in the absence of PHS, the Probability of *QSFD* recovery is just 14.67 %, substantially lower than when PHS is involved. With FS-PHS and VS-PHS, this probability increases to 53.52 % and 61.76 %, respectively. This difference is mainly due to insufficient system inertia and regulation capacity without PHS, limiting frequency recovery. In contrast, VS-PHS, with its superior flexibility in power regulation, offers a more significant improvement in frequency recovery compared to FS- PHS. Additionally, in both SCUC and FBUC modes, PHS participation enhances system frequency recovery performance, with *QSFD* values predominantly below the 0.2 Hz threshold and concentrated around 0.15 Hz. Notably, VS-PHS reaches a 100 % recovery probability, outperforming FS-PHS by 3 %–5 %, owing to its faster response and broader power regulation range.

5.2.2. Delivery reliability assessment

Fig. 17 presents the distribution of flexibility insufficiency probability (*FIP*) across TCUC, SCUC, and FBUC operating models, including both upward flexibility insufficiency probability (*UFIP*) and downward flexibility insufficiency probability (*DFIP*), as well as the renewable energy curtailment rate (*ECR*) and power shortage rate (*PSR*) resulting from flexibility insufficiency. These metrics collectively assess the CHIDS's reliability during the delivery process and underscore the contribution of PHS in enhancing flexibility supply and improving system reliability.

As presented in Fig. 17, in the absence of PHS, the *DFIP* in the TCUC mode is the highest, reaching 21.73 %, with a *ECR* of 13.96 %. This phenomenon exposes the inherent limitations of conventional economic dispatch strategies—namely, the insufficient downward regulation capacity that compels the curtailment of a substantial volume of VRE. Notably, the integration of PHS results in a marked reduction in the curtailment rate by over 10 % within the TCUC model, thereby substantiating the bidirectional regulation capability of PHS in enhancing power delivery flexibility.

Both the SCUC and FBUC models, through the incorporation of frequency security constraints, markedly improve the system's dynamic response; however, this improvement is accompanied by a trade-off in flexibility. Under the SCUC model, Case 1—with FS-PHS—exhibits increases in *DFIP* and *UFIP* to 19.19 % and 33.07 %, respectively, while the *ECR* and *PSR* values escalate to 5.97 % and 3.35 %. In contrast, Case 2—with VS-PHS—records *DFIP* and *UFIP* values of 17.41 % and 19.49 %, respectively, with corresponding *ECR* and *PSR* of 4.52 % and 1.03 %. This conservative configuration primarily arises from the inflexible reserve capacity requirements imposed by the frequency security constraints, which necessitate the allocation of certain flexibility resources exclusively for frequency support. Conversely, the FBUC model achieves a substantial reduction in the probability of flexibility inadequacy through the dynamic optimization of reserve allocation. Specifically, in Case 2 employing VS-PHS, *DFIP* and *UFIP* are reduced to 9.53 % and 10.35 %, respectively, with the curtailment and shortage rates decreasing to 4.29 % and 0.7 %. Relative to Case 1, these adjustments result in curtailment and shortage rate reductions of 0.78 % and 0.55 %, respectively.

5.2.3. Operational economic assessment

Fig. 18 presents a comparative analysis of the economic evaluation results under the TCUC, SCUC, and FBUC operating models, highlighting the significant role of PHS in enhancing the economic efficiency of the CHIDS from multiple perspectives.

Fig. 18(a) presents the distribution of peak-shaving reserve capacity in PHS and cascade hydropower plants under pumping and generation conditions across TCUC, SCUC, and FBUC operating modes. Given the critical importance of upward flexibility for power delivery reliability, this section emphasizes the analysis of upward flexibility supply. The results demonstrate that VS-PHS plays a key role in providing reserve capacity, particularly in the FBUC mode, where it contributes approximately 67 % of the total reserve capacity. In contrast, FS-PHS units, with their fixed output characteristics, provide the least flexibility, contributing only around 11 % to the peak shaving supply. This highlights the flexible peak-shaving capability of VS-PHS, enhancing the overall reliability and economics of the system.

Fig. 18(b) further illustrates the distribution of primary and secondary frequency regulation reserve capacities in the system under the FS-PHS participation case in the FBUC mode, considering both pumping and generation conditions. The cumulative annual supply of primary and secondary frequency regulation from the CHIDS is approximately 420 GW and 2964 GW, respectively. While the PFR capability of PHS is

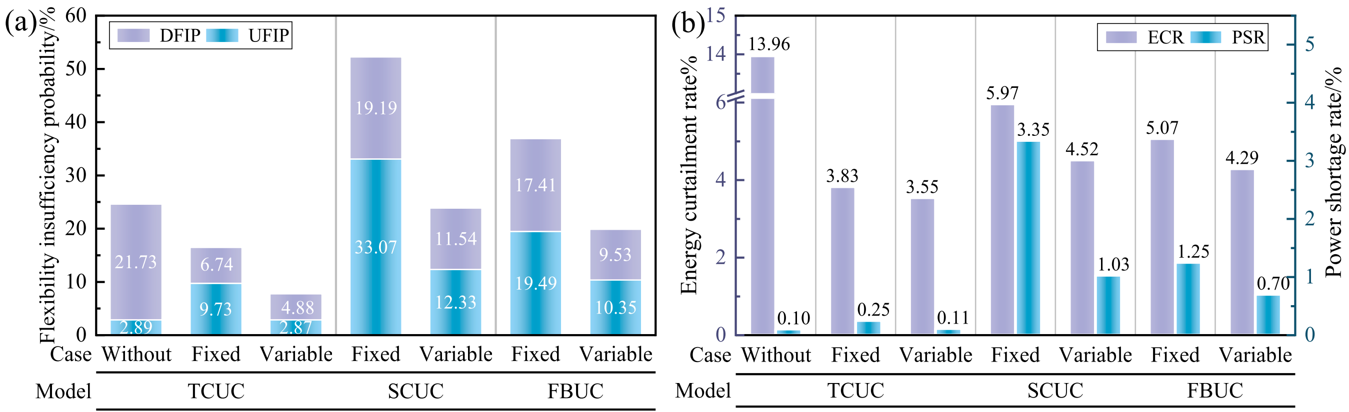


Fig. 17. Comparative analysis of flexibility insufficiency probability under different operating scenarios (a), and the renewable energy curtailment rate and power shortage rate (b).

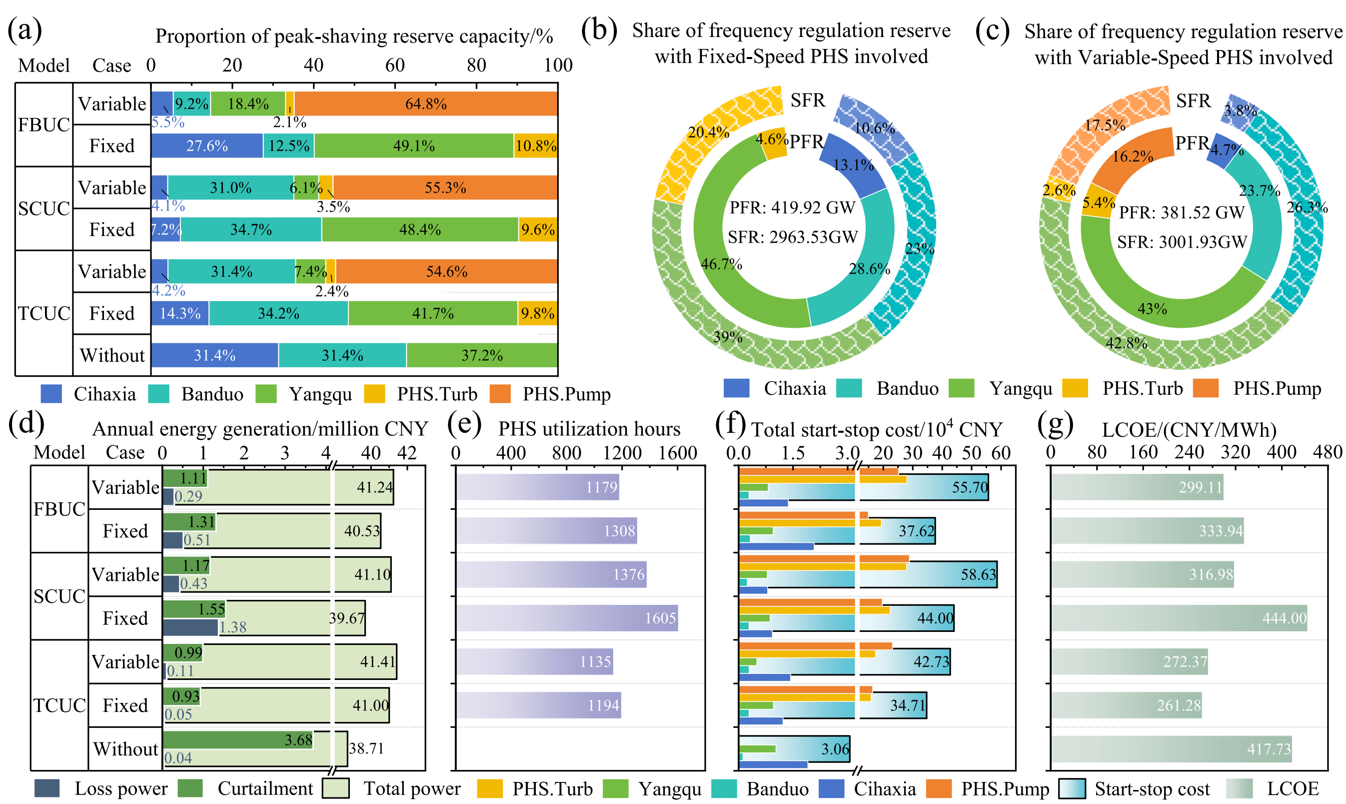


Fig. 18. Comparative analysis of economic evaluation metrics for the CHIDS under different operating scenarios. (a) Proportion of peak-shaving reserve capacity, (b) Share of frequency regulation reserve with fixed-speed PHS involved, (c) Share of frequency regulation reserve with variable-speed PHS involved, (d) Annual energy generation, (e) Effective utilization hours of PHS (EUR_{PHS}), (f) Total start-up and shut-down costs, (f) Levelized cost of electricity (*LCOE*), and (g) Internal rate of return (*IRR*).

relatively limited in the generation mode (4.6 %), its contribution to SFR is substantial, at 20.4 %. This highlights the critical role of PHS in SFR, particularly in mitigating long-duration frequency fluctuations and compensating for the limitations of conventional units during frequency restoration, thereby enhancing overall frequency regulation performance.

Fig. 18(c) illustrates the distribution of reserve capacities for primary and secondary frequency regulation under the VS-PHS participation case in the FBUC mode. The system provides a cumulative annual supply of approximately 382 GW for PFR reserve and 3002 GW for SFR reserve. Compared to fixed-speed units, variable-speed units offer greater flexibility in both pumping and generation conditions, particularly by adjusting the head difference and controlling flow to provide upward regulation. In this scenario, PHS contributes 5.4 % and 16.2 % to primary frequency regulation in pumping and generation modes, respectively, and 2.6 % and 17.5 % to secondary regulation. Notably, PHS provides significantly lower upward regulation reserve in generation mode due to its load-following role, which limits rapid response to frequency fluctuations.

Fig. 18(d) illustrates the system's power delivery under different scenarios across the three operating models. The results demonstrate that greater participation of PHS significantly reduces curtailment, particularly in the FBUC mode, where curtailment is substantially lower than in SCUC. This indicates that the FBUC mode enhances the utilization of VRE by optimizing scheduling, while alleviating the curtailment pressure induced by the volatility and uncertainty of VRE. Although annual generation in FBUC is slightly lower than in TCUC, it exceeds that of SCUC, suggesting a more favorable balance between economic efficiency and system flexibility.

As shown in Fig. 18(e), under the frequency security constraints, the annual effective operating hours of PHS significantly increase, especially in the SCUC mode where the utilization is maximized. This demonstrates that incorporating frequency security constraints optimizes PHS operational strategy, improving its efficiency, economic value, and scheduling flexibility within the CHIDS.

Fig. 18(f) presents the start-up and shut-down costs of cascade hydropower stations and PHS plants under pumping and generation conditions across various operating models. The results indicate that PHS start-up and shut-down costs are consistently higher than those of cascade hydropower stations, primarily due to the greater number of PHS units and additional losses from frequent mode switching. Furthermore, the total start-up and shut-down costs in the FBUC mode are lower than in the SCUC mode, highlighting the optimization benefits of flexible scheduling. Specifically, for both fixed-speed and variable-speed units, start-up and shut-down costs in the FBUC mode are 14.5 % and 5 % lower than in SCUC, respectively. Additionally, the start-up and shut-down costs of fixed-speed units in FBUC are 32 % lower than those of variable-speed units, demonstrating the cost-reducing potential of the proposed optimization scheduling method.

The *LCOE* is a critical metric for evaluating the cost of power generation across different operating scenarios, as shown in Fig. 18(g). In the case of FS-PHS, the *LCOE* of CHIDS in the SCUC mode is 444.0 CNY/MWh, which is 26.3 CNY/MWh higher than the baseline scenario without PHS participation and using the TCUC mode. This suggests that transient frequency security constraints alone do not significantly enhance system economics, as the added flexibility requirements increase start-up frequencies and reserve capacity, thereby raising operational costs. Additionally, the SCUC mode does not fully leverage the flexible regulation capabilities of PHS. In contrast, in the FBUC mode with FS-PHS, the *LCOE* decreases to 333.94 CNY/MWh, representing reductions of 20.1 % and 24.8 % compared to the baseline and SCUC models, respectively. This improvement is attributed to the FBUC mode, which incorporates frequency regulation reserve costs into the optimization objective while accounting for steady-state power regulation, transient frequency response, and quasi-steady-state frequency recovery constraints, thus enhancing PHS operational efficiency. For the VS-PHS case, the *LCOE* in FBUC is 299.11 CNY/MWh, 5.6 % lower than in SCUC, reflecting the superior dynamic regulation capabilities of VS-PHS, which enhance frequency response efficiency and optimize system flexibility.

5.3. Optimal for VRE integration capacity based on flexibility enhancement metrics

This section builds upon the VRE integration capacity assessment method introduced in Section 3, combining system operational characteristics with flexibility enhancement evaluation metrics to explore the dual optimization of VRE in terms of both system frequency security and operational economics. Fig. 19 illustrates the impact of variations in renewable energy penetration on flexibility enhancement indicators under the TCUC and FBUC operational modes, as demonstrated in cases involving both FS-PHS and VS-PHS units.

Regarding frequency security, as shown in Fig. 19(a), under the TCUC mode, as VRE penetration increases from 47 % to 74 %, the VD_{RoCoF} exhibits a monotonically decreasing trend. However, its values remain above zero, indicating that the frequency safety margin continues to be compromised. Simultaneously, the RP_{QSFR} displays a nonlinear upward trend with increasing penetration, peaking at 82 % when penetration reaches 62 %, after which it saturates due to a reduction in system inertia. This phenomenon underscores an intrinsic contradiction in conventional economic dispatch: while increased VRE penetration bolsters the supply of steady-state flexibility, it concurrently diminishes the system's dynamic frequency response capability. In

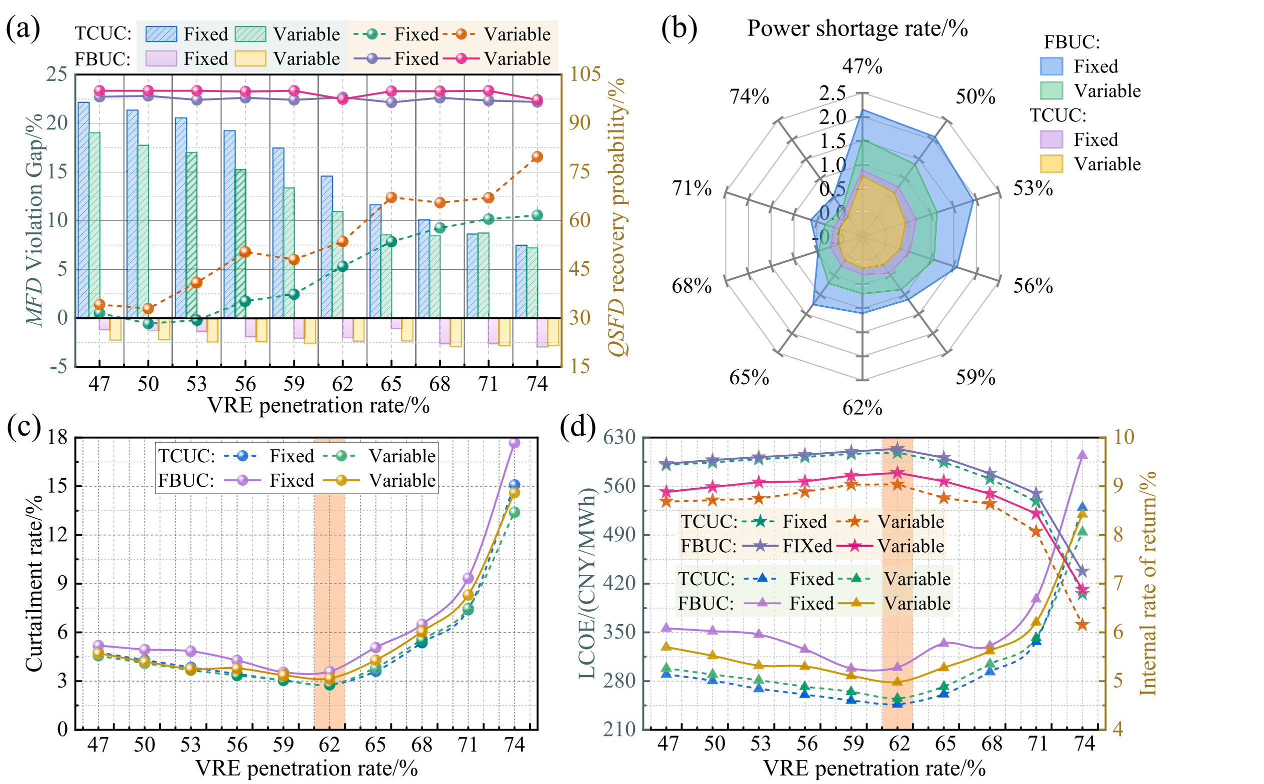


Fig. 19. Impact of VRE penetration on flexibility enhancement evaluation metrics for TCUC and FBUC operational models. (a) Frequency response security, (b) Power shortage rate, (c) Renewable energy curtailment rate, and (d) Operational economics.

contrast, under the FBUC mode, VD_{RoCoF} ultimately attains negative values and RP_{QSFR} consistently remains around 98 % with fluctuations confined within ± 2 %, thereby validating the robustness of this mode in high-penetration scenarios. Notably, when renewable penetration reaches 65 %, the quasi-steady-state frequency deviation recovery probability in cases involving VS-PHS units is enhanced by 3.4 % compared to FS-PHS cases, primarily attributable to the rapid power modulation capabilities of VS-PHS units that effectively compensate for the degradation of system inertia.

In the analysis of delivery reliability, as depicted in Fig. 19(b), the TCUC mode maintains the system's power deficiency rate below 1 %, ensuring power delivery reliability exceeding 99 %. However, this elevated reliability is achieved at the expense of frequency security. Specifically, the TCUC mode maximizes the utilization of transmission corridor capacity to prioritize delivery power supply, which results in insufficient system inertia support and a significantly heightened risk of frequency instability. Conversely, the FBUC mode, through a dynamic optimization strategy for reserve capacity allocation, restricts the power shortage rate to within 2.15 % when VRE penetration is below 56 %, and further reduces it to under 1 % when penetration exceeds 65 %. It is noteworthy that under the FBUC mode, when VRE penetration lies within the 50 %–70 % range, the power shortage rate in cases involving VS-PHS is reduced by 0.25–0.79 % relative to FS-PHS cases, owing predominantly to the enhanced system flexibility conferred by the rapid power response and wide-range adjustment capabilities of variable-speed units.

Further observation of Fig. 19(c) reveals that, under both TCUC and FBUC models, as VRE penetration rises, the curtailment rate exhibits a trend characterized by an initial gradual decline followed by a rapid escalation. Specifically, at lower penetration levels (<62 %), the curtailment rate decreases from 5.2 % to 3.0 %, a phenomenon mainly attributable to the spatiotemporal complementary regulation between PHS and cascaded hydropower stations, which significantly enhances transmission line utilization and effectively alleviates VRE integration pressure. However, at higher penetration levels (>62 %), the system becomes constrained by transmission capacity and the modulation margin of flexibility resources, and in conjunction with the reverse peak-shaving effect of VRE output, the CHIDS is compelled to reduce output to maintain power balance, thereby inducing an exponential increase in the curtailment rate.

From an economic standpoint, Fig. 19(d) demonstrates that, the system's *LCOE* initially declines with increasing VRE penetration and subsequently rises, reaching a minimum at 62 %. In the FBUC mode, the *LCOE* for FS-PHS and VS-PHS participation is 299.5 CNY/MWh and 278.4 CNY/MWh, respectively, reflecting increases of 21.4 % and 9.1 % compared to the TCPU mode. This suggests that while the FBUC mode improves frequency security and delivery reliability, it incurs slightly higher operational costs due to the increased flexibility. Additionally, the IRR follows a similar trend to *LCOE*, peaking at 62 % VRE penetration. The *IRR* for FS-PHS and VS-PHS participation case in the FBUC mode is 9.8 % and 9.3 %, showing increases of 0.08 % and 0.24 % over the TCUC mode. These results indicate that the FBUC mode can effectively balance the trade-off between flexibility, costs, and economic efficiency.

Based on the above analysis, when VRE penetration reaches 62 %, the system exhibits optimal performance across all flexibility-enhancement metrics: frequency security complies with operational requirements, power delivery reliability approaches 99 %, both the curtailment rate and *LCOE* are minimized, and the *IRR* is maximized. This critical point reflects the dynamic equilibrium threshold between the provision of system flexibility and the variability demand of renewable energy. Beyond this threshold, the modulation margin of PHS becomes saturated, and the reverse peak-shaving effect intensifies, leading to a nonlinear escalation in both the curtailment rate and operational costs.

6. Conclusions

To resolve the inherent conflict between the flexible regulation capability of PHS and the economic operation of high-penetration VRE systems, this study introduces a multidimensional flexibility balancing constraint framework. This framework integrates three pivotal dimensions, namely steady-state power regulation, transient frequency response, and quasi-steady state recovery, and develops a multi-timescale scheduling model (FBUC) based on nested day-ahead and intraday optimization. Simultaneously, a decision-making methodology for VRE integration capacity is proposed, which concurrently considers both economic efficiency and transient security.

Empirical investigations conducted on the CHIDS in Qinghai Province, China, demonstrate that, compared with the traditional SCUC approach, the FBUC model under various scheduling scenarios involving FS-PHS and VS-PHS reduces the probability of insufficient system flexibility by 15 % and 4 %, respectively, while decreasing unit startup and shutdown costs by 14.5 % and 5 %. With respect to frequency security, the proposed model maintains a quasi-steady state frequency deviation recovery probability of up to 98 %. Moreover, by incorporating the rapid power-tracking capability of VS-PHS, the probability of power deficiency during external transmission is further reduced by 0.25 % to 0.79 % compared to FS-PHS. When VRE penetration reaches 62 %, the overall system performance is optimized, achieving an external transmission reliability approaching 99 % and a curtailment rate as low as 3 %, with internal rates of return of 9.8 % for FS-PHS and 9.3 % for VS-PHS—improvements of 0.08 % and 0.24 %, respectively, over traditional scheduling strategies that neglect frequency constraints (TCUC).

The study also reveals a significant functional complementarity between the two types of PHS units, with FS-PHS exhibiting marked cost advantages in base load regulation, whereas VS-PHS, owing to its superior transient response capability, is more proficient at accommodating abrupt fluctuations in VRE output. In view of these distinct operational strengths, the proposed framework can be effectively extended to mixed deployment scenarios of FS-PHS and VS-PHS. Future research should further investigate the synergistic enhancement mechanisms inherent in mixed configurations, thereby elucidating the optimal complementary capacity ratio between the two unit types.

It is noteworthy that the current model is predicated on an assumption of constant efficiency, which does not fully capture the nonlinear head–efficiency characteristics that affect unit flexibility. In particular, the differential efficiency degradation of FS-PHS under low head conditions, in contrast to the sustained efficiency performance of VS-PHS, has not been quantitatively assessed. Future work should focus on developing a refined model that more accurately represents these dynamics, thereby further enhancing the evaluation framework.

CRediT authorship contribution statement

Xingjin Zhang: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Jizhou Wang:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Ziwen Zhao:** Writing – review & editing, Methodology, Conceptualization. **Fangfang Li:** Writing – review & editing, Supervision, Funding acquisition. **Pin Jern Ker:** Writing – review & editing, Formal analysis. **Ali Najah Ahmed:** Writing – review & editing, Validation, Supervision. **Beibei Xu:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Diyi Chen:** Writing – review & editing, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Output modeling

A.1. Modeling of the output of cascade hydropower stations

(1) Hydraulic connection constraint.

$$I_{i,t}^{HP} = Q_{i-1,t-\tau}^{HP} + QI_{i,t} \tag{A.1}$$

where $I_{i,t}$ is the inflow to reservoir i at time t, τ is the hydraulic delay between stations i - 1 and i, $Q_{i-1,t-\tau}$ is the outflow of station i - 1 at time t - τ; $QI_{i,t}$ is the interval runoff between stations i - 1 and i.

(2) Water balance constraint of the reservoir.

$$V_{i,t}^{HP} = V_{i,t-1}^{HP} + \left(I_{i,t-1}^{HP} - Q_{i,t}^{HP} - Q_{i,t}^{spill}\right)\cdot\Delta t \tag{A.2}$$

where $V_{i,t}$ is the reservoir volume, $Q_{i,t}^{HP}$ and $Q_{i,t}^{spill}$ are the penstock release and spillage from the reservoir i at time t, respectively.

(3) Outflow constraints.

$$\begin{cases} Q_{i,t}^{HP} = \sum_{g\in N_{HP,i}} q_{i,g,t}^{HP} \\ Q_{i,\min}^{HP} \le Q_{i,t}^{HP} \le Q_{i,\max}^{HP} \\ 0 \le Q_{i,t}^{spill} \le Q_{i,\max}^{spill} \end{cases} \tag{A.3}$$

where $Q_{i,min}^{HP}$ and $Q_{i,max}^{HP}$ are the minimum and maximum generation flow of station i during the period t, respectively. $q_{i,g,t}^{HP}$ is the generation flow of hydropower unit g at time t.

(4) Reservoir characteristic constraint.

$$\begin{cases} Z_{i,t}^{HP} = f_i^{ZV}\left(V_{i,t}^{HP}\right) \\ Z_{i,t}^{tail} = f_i^{ZQ}\left(Q_{i,t}^{HP} + Q_{i,t}^{spill}\right) \end{cases} \tag{A.4}$$

where $Z_{i,t}^{HP}$ and $Z_{i,t}^{tail}$ are the fore-bay water level and tailrace water level, respectively. f_i^{ZV} and f_i^{ZQ} are the reservoir storage capacity curve and tailwater rating curve, respectively.

(5) Scheduling start-end boundary constraint.

$$\begin{cases} Z_{i,\min}^{HP} \le Z_{i,t}^{HP} \le Z_{i,\max}^{HP} \\ Z_{i,0}^{HP} = Z_{i,ini}^{HP} \\ |Z_{i,T}^{HP} - Z_{i,end}^{HP}| \le \Delta Z_i^{HP} \end{cases} \tag{A.5}$$

where $Z_{i,min}^{HP}$ and $Z_{i,max}^{HP}$ denote the upper and lower limits of the reservoir water level, respectively. $Z_{i,ini}^{HP}$, $Z_{i,end}^{HP}$, and ΔZ_i^{HP} represent the initial water level, final water level, and the permissible variation in the final water level for the reservoir scheduling, respectively.

(6) Hydropower unit output characteristic.

$$P_{i,g,t}^{HP} = 9.81\rho_w \eta_{i,g}^{HP} q_{i,g,t}^{HP} H_{i,g,t} \tag{A.6}$$

$$H_{i,g,t} = \left(Z_{i,t-1}^{HP} + Z_{i,t}^{HP}\right)\Big/2 - Z_{i,t}^{tail} - H_{loss} \tag{A.7}$$

where ρ_w is the water density, g is the gravitational acceleration, $\eta_{i,g}^{HP}$ is the turbine efficiency, $H_{i,g,t}$ is net hydraulic head, H_{loss} is the head loss.

(7) Hydraulic head constraint of hydropower units.

$$H_{i,g,\min} \leq H_{i,g,t} \leq H_{i,g,\max} \tag{A.8}$$

where $H_{i,g,\min}$ and $H_{i,g,\max}$ are the lower and upper hydraulic head limits for unit g, respectively.

(8) Output limitation of hydropower units.

$$u^{HP}_{i,g,t} P^{HP}_{i,g,\min} \leq P^{HP}_{i,g,t} \leq u^{HP}_{i,g,t} P^{HP}_{i,g,\max} \tag{A.9}$$

where $P^{HP}_{i,g,t}$ represents the power output of unit g, $P^{HP}_{i,g,min}$ and $P^{HP}_{i,g,max}$ indicate the minimum and maximum output limits, $u^{HP}_{i,g,t}$ is the unit's operational state variable at time t.

(9) Vibration zone traversal constraint for units.

$$\left(P^{HP}_{i,g,t} - P^{V,\max}_{i,g,m}\right)\left(P^{HP}_{i,g,t} - P^{V,\min}_{i,g,m}\right) \geq 0 \tag{A.10}$$

where $P^{V,max}_{i,g,m}$ and $P^{V,min}_{i,g,m}$ are the lower and upper output limits of unit g in the m^{th} vibration zone.

(10) Minimum start-up and shutdown time constraint.

$$S^{HP}_{i,g,t} = u^{HP}_{i,g,t} - u^{HP}_{i,g,t-1}$$
$$\begin{cases} S^{HP}_{i,g,t} - S^{HP}_{i,g,t+\theta} \leq 1, \forall \theta \in \left(1, T_{on,i,g}\right) \\ S^{HP}_{i,g,t} - S^{HP}_{i,g,t+\theta} \geq -1, \forall \theta \in \left(1, T_{off,i,g}\right) \end{cases} \tag{A.11}$$

where the state variable $S^{HP}_{i,g,t}$ takes values of 0, 1, or -1, corresponding to no change, start-up, or shutdown of the unit relative to the previous time period. $T_{on,i,g}$ and $T_{off,i,g}$ denote the minimum operating and shutdown times for unit g in hydropower station i, respectively.

A.2. Wind power output model

The power output of wind turbines is directly influenced by wind speed variations. The specific power output model is as follows:

$$P^{WT}_t = \begin{cases} 0, & v_t < v_{in} \text{ or } v_t > v_{out} \\ \dfrac{v_t^3 - v_{in}^3}{v_r^3 - v_{in}^3} \cdot P^{WT}_r, & v_{in} \leq v_t \leq v_r \\ P^{WTs}_r, & v_r \leq v_t \leq v_{out} \end{cases} \tag{A.12}$$

where *PWT r* denotes the rated power of the wind turbine, v_{in}, v_r and v_{out} represent the cut-in, rated, and cut-out wind speeds of the turbine, respectively. and v_t is the wind speed at time t.

A.3. Photovoltaic power output model

The power output of PV systems is affected by solar irradiance and ambient temperature, displaying distinct diurnal periodicity. The power output model is as follows:

$$P^{PV}_t = A_{PV} \cdot \eta_{PV} \cdot G_t \cdot \left[1 - \theta_{Tc} \cdot \left(T_{c,t} - T_{ref}\right)\right] \tag{A.13}$$

where *PPV t* represents the PV output at time t, A_{PV} is the PV array area, η_{PV} denotes the PV conversion efficiency, G_t is the solar irradiance at time t, θ_{TC} is the temperature coefficient, $T_{c,t}$ denotes the operating temperature, T_{ref} is the reference temperature under standard conditions.

A.4. Operation model of pumped hydro storage system

(1) Water balance constraint for reservoirs.

$$V^{up}_t = V^{up}_{t-1} + \left(\sum_{g=1}^{N_{PS}} q^{PS.P}_{g,t} - \sum_{g=1}^{N_{PS}} q^{PS.G}_{g,t}\right) \cdot \Delta t \tag{A.14}$$

$$V^{lw}_t = V^{lw}_{t-1} + \left(\sum_{g=1}^{N_{PS}} q^{PS.G}_{g,t} - \sum_{g=1}^{N_{PS}} q^{PS.P}_{g,t}\right) \cdot \Delta t \tag{A.15}$$

where V^{lw}_t is the downstream reservoir's water volume at time t. $q^{PS.G}_{g,t}$ and $q^{PS.P}_{g,t}$ are the flow rates of PHS unit g in generation and pumping modes, respectively.

(2) Reservoir capacity constraints.

$$\begin{cases} V_{\min}^{up} \le V_t^{up} \le V_{\max}^{up} \\ V_{\min}^{lw} \le V_t^{lw} \le V_{\max}^{lw} \end{cases} \tag{A.16}$$

where V_{min}^{up} and V_{max}^{up} are the lower and upper capacities of the upstream reservoir, respectively. V_{min}^{lw} and $V_{max}^{lw}$are the lower and upper capacities of the downstream reservoir.

(3) Initial and final capacity constraints of the upstream reservoir.

$$\begin{cases} V_0^{up} = V_{initial}^{up} \\ V_T^{up} = V_{end}^{up} \end{cases} \tag{A.17}$$

where V_{ini}^{up} and V_{end}^{up} represent the initial and final capacities of the upstream reservoir.

(4) Conversion efficiency of power output and input flow.

$$\begin{cases} q_{g,t}^{PS.G} = \xi_g^{PS.G} P_{g,t}^{PS.G}, \xi_g^{PS.G} = \dfrac{1}{9.81\rho_w \eta_g^{PS.G} H_{ave}} \\ q_{g,t}^{PS.P} = \xi_g^{PS.P} P_{g,t}^{PS.P}, \xi_g^{PS.P} = \dfrac{\eta_g^{PS.P}}{9.81\rho_w H_{ave}} \end{cases} \tag{A.18}$$

where $\xi_g^{PS.G}$ and $\xi_g^{PS.P}$ denote the hydraulic conversion coefficients for unit g in generation and pumping modes, respectively. $\eta_g^{PS.G}$ and $\eta_g^{PS.P}$ the generation and pumping efficiencies, respectively. H_{ave} denotes the average hydraulic head.

(5) Power output limitations in pumping and generation modes.

$$u_{g,t}^{PS.G} P_{\min,g,t}^{PS.G} \le P_{g,t}^{PS.G} \le u_{g,t}^{PS.G} P_{\max,g,t}^{PS.G} \tag{A.19}$$

$$u_{g,t}^{PS.P} P_{g,\min}^{PS.P} \le P_{g,t}^{PS.P} \le u_{g,t}^{PS.P} P_{g,\max}^{PS.P} \tag{A.20}$$

(6) Operation mode transition constraint.

$$0 \le u_{g,t}^{PS.G} + u_{g,t}^{PS.P} \le 1 \tag{A.21}$$

(7) Start-stop frequency constraint.

$$\sum_{t\in T} \begin{bmatrix} u_{g,t}^{PS,G}\left(1 - u_{g,t-1}^{PS,G}\right) + u_{g,t-1}^{PS,G}\left(1 - u_{g,t}^{PS,G}\right) \\ +u_{g,t}^{PS,P}\left(1 - u_{g,t-1}^{PS,P}\right) + u_{g,t-1}^{PS,P}\left(1 - u_{g,t}^{PS,P}\right) \end{bmatrix} \le M_{PS,g} \tag{A.22}$$

where $M_{PS,g}$ denotes the maximum allowable number of start-stop cycles for unit g.

(8) Minimum start-stop time constraint.

$$\begin{cases} S_{g,t}^{PS.G} = u_{g,t}^{PS.G} - u_{g,t-1}^{PS.G} \\ S_{g,t}^{PS.G} - S_{g,t+\theta}^{PS.G} \le 1, \ \forall \theta \in \left(1, T_{on,g}^{PS}\right) \\ S_{g,t}^{PS.G} - S_{g,t+\theta}^{PS.G} \ge -1, \forall \theta \in \left(1, T_{off,g}^{PS}\right) \end{cases} \tag{A.23}$$

$$\begin{cases} S_{g,t}^{PS.P} = u_{g,t}^{PS.P} - u_{g,t-1}^{PS.P} \\ S_{g,t}^{PS.P} - S_{g,t+\theta}^{PS.P} \le 1, \ \forall \theta \in \left(1, T_{on,g}^{PS}\right) \\ S_{g,t}^{PS.P} - S_{g,t+\theta}^{PS.P} \ge -1, \forall \theta \in \left(1, T_{off,g}^{PS}\right) \end{cases} \tag{A.24}$$

where $S_{g,t}^{PS.G}$ and $S_{g,t}^{PS.P}$ indicate the unit's operational status under generation and pumping modes, respectively. A value of 0 signifies an unchanged status relative to the previous period, whereas 1 and -1 denote initiation and termination, respectively. $T^{PS}_{on,g}$ and $T^{PS}_{off,g}$ are the minimum continuous operating time and shutdown duration for PHS unit g.

Appendix B. Frequency response model of hydropower units

The frequency response of the hydro units' regulation system primarily stems from synchronous inertia, turbine, and governor control loops [51]. The typical structural model for frequency regulation of hydro units is depicted in Fig. B.1, where H^{HP} is inertia time constant, T_G is the governor time constant, R_P is permanent difference coefficient, R_C is the temporary droop constant, T_d represents the reset time constant, T_w denotes the water flow inertia time constant.

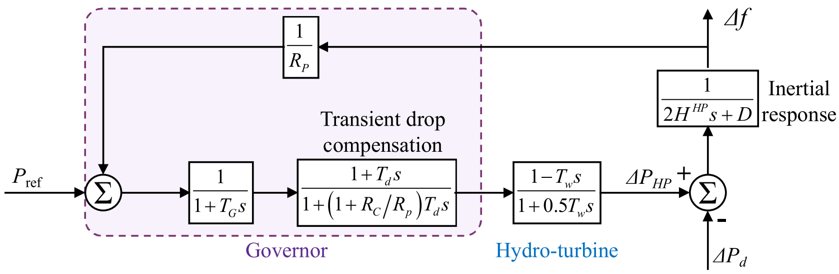

Fig. B.1. Frequency characteristic block diagram of hydropower units [52].

The analysis from Fig. B.1 shows that the frequency response model of hydropower units is a high-order model, which hampers the dynamic frequency analysis of the CHIDS. Hence, the transfer function of the frequency response is approximated using a transient droop compensation link [53], as specified in Eq. (B.1).

$$G_{HP}(s) = \frac{\Delta P_{HP}(s)}{\Delta f(s)} = \frac{1}{R_p}\cdot\frac{1 + T_d s}{1 + R_C T_d s} \tag{B.1}$$

Appendix C. Derivation of frequency deviation extremum

Maximum frequency deviation (Δf_{nadir}) is derived from the frequency response time-domain model of the CHIDS. However, from Fig. 7, the structural parameters of the governor of various units within the system are not identical, posing challenges in transforming the system's frequency response model into a time-domain analytical expression, thus hindering the determination of the Δf_{nadir}. Hence, utilizing the standardized frequency control structure in Fig. C.1 [42], the frequency response parameters of diverse unit types within the CHIDS are transformed into standard governor parameters via the conversion method outlined in Eq. (C.1).

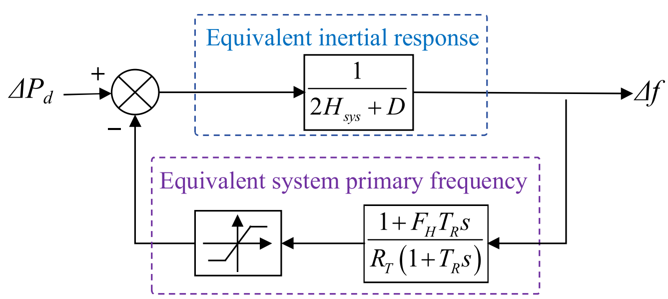

Fig. C.1. Equivalent frequency response model of the integrated delivery system.

$$\begin{cases} F_H^{HP} = 1/R_C, R_T^{HP} = R_p, T_R^{HP} = R_C T_d \\ F_H^{PS} = \dfrac{K_{PS}^P}{1 + K_{PS}^P}, R_T^{PS} = e_p, T_R^{PS} = \dfrac{1 + K_{PS}^P}{K_{PS}^I} \end{cases} \tag{C.1}$$

where $Fx\,H$ is the fraction of total power generated by the turbine, $Tx\,R$ is the reheat time constant, $Rx\,T$ is droop gain. $x \in \{HP, PS\}$, indicates the type of power station.

The dynamic frequency response model of the CHIDS after equivalence at time t still contains numerous parameters, which can be aggregated to reduce computational complexity. The process of aggregation is detailed in Eqs. (C.2)-(C.5) [54].

$$\kappa_{g,t} = \frac{P_{g,\max}}{Cap_{sys}} u_{g,t}, \quad \forall g \in N_u \tag{C.2}$$

$$F_{H,t} = \sum_{g \in N_u} \frac{F_{H,g} \cdot \kappa_{g,t}}{R_{T,g}}, \forall g \in N_u \tag{C.3}$$

$$R_{T,t} = \sum_{g \in N_u} \frac{\kappa_{g,t}}{R_{T,g}}, \forall g \in N_u \tag{C.4}$$

$$T_{R,t} = \sum_{g \in N_u} \frac{T_{R,g} \cdot \kappa_{g,t}}{R_{T,g}}, \forall g \in N_u \tag{C.5}$$

Constraints (C2-C5) calculate equivalent system variables for power fraction, droop factor, time constant, respectively. $\kappa_{g,t}$ is the g^{th} unit scaled power gain factor at time t, $P_{g,\max}$ is the maximum output power of the g^{th} unit with frequency modulation function.

Subsequently, the equation for the Δf_{nadir} of CHIDS after a step disturbance can be analytically derived as Eq. (C.6), the detailed derivation can be referred to in [42].

$$\Delta f_{ndair} = \frac{|\Delta P_d|}{D + R_T}\left(1 + \sqrt{\frac{T_R(R_T - F_H)}{2H_{sys}}} e^{-\xi \omega_n t_{nadir}}\right) \tag{C.6}$$

$$\begin{cases} \omega_n = \sqrt{\dfrac{D + R_T}{2H_{sys} \cdot T_R}}, \omega_r = \omega_n \sqrt{1 - \xi^2} \\ \xi = \dfrac{2H_{sys} + T_R \cdot (D + F_H)}{2\sqrt{2H_{sys} \cdot T_R \cdot (D + R_H)}} \\ t_{nadir} = \dfrac{1}{\omega_r} \tan^{-1}\left(\dfrac{\omega_r}{\xi \cdot \omega_n - 1/T_R}\right) \end{cases} \tag{C.7}$$

where ω_n is the natural frequency, ξ is the damping ratio, t_{naidr} is the time instance of frequency nadir.

Appendix D. Model input data

Table D1
Specific parameters of hydropower stations.

Type	Parameters	Symbol	Unit	Cihaxia	Banduo	Yangqu
Reservoir	Regulated storage capacity	V^{HP}	10^7 m^3	185.4	0.196	23.9
	Normal pool level	Z^{HP}_{max}	m	2990	2760	2715
	Dead water level	Z^{HP}_{min}	m	2975	2757	2710
	Minimum ecological discharge	Q^{HP}_{min}	m^3/s	100	100	100
Hydroelectric generator	Number of hydro units	N_{HP}	p.u.	4	3	3
	Minimum output	P^{HP}_{max}	MW	650	120	400
	Minimum technical output	P^{HP}_{min}	MW	130	24	80
	Ramping speed	ΔP^{HP}_{ramp}	MW/min	20	4	13
	Efficiency	η^{HP}	%	90.65	87.0	88.0
	Maximum discharge	q^{HP}_{max}	m^3/s	342.25	373.16	395.4
	Minimum operating/shutdown time	T_{on}/ T_{off}	min	60	30	45
	Inertial time constant	H_{HP}	second	10	6	8
Governor	Permanent difference coefficient	R_P	p.u.	0.04	0.04	0.04
	Temporary droop	R_C	p.u.	0.36	0.43	0.667
	Reset time	T_d	s	6	3.38	5
Cost	Startup and shutdown cost	C^{HP}_{on}	CNY	60	12	35
	Peak-shaving reserve cost	C^{HP}_{peak}	CNY/MWh	2.4	1.6	2.0
	PFR reserve cost	C^{HP}_{PFR}	CNY/MWh	4.3	2.3	2.6
	SFR reserve cost	C^{HP}_{SFR}	CNY/MWh	2.5	1.8	2.3

Appendix E. Nomenclature

A. **Abbreviations**		*KI PS, KP PS*	Integral gain and proportional gain.
CHIDS	Cascade hydro-wind-solar-storage integrated delivery system	*M*	Sufficiently large positive number
ECR	Renewable energy curtailment rate	$N_{HP,i}$	Number of hydropower units in station *i*.
EUR_{PHS}	Annual effective utilization hours of PHS	N_{PS}	Number of PHS units.
FBUC	Flexibility-balanced unit commitment	p_k	Probability of scenario *k*.
FIP	Flexibility insufficiency probability	*PDC k,t*	Delivery power at time *t* under scenario *k*.
FS-PHS	Fixed-speed pumped hydro storage	*PPS.G g,t, PPS·P g,t*	Generation and pumping power of PHS.
IRR	Internal rate of return	*PHP i,g,t*	Power output of hydro unit *g* in station *i*.
LCOE	Levelized cost of electricity	*PWT f,t, PPV f,t*	Predicted output of wind and PV power.
MFD	Maximum frequency deviation	*PWT k,t, PPV k,t*	Actual output at time *t* under scenario *k*.
MILP	Mixed-integer linear programming	*Pspill k,t, Pcur k,t*	Curtailment of hydro and VRE.
NPV	Net present value	*Ploss k,t*	Power shortage at time *t* under scenario *k*.
PFR	Primary frequency regulation	*PLoad k,t*	Load demand at the receiving-end grid.

(continued on next page)

(*continued*)

PHS	Pumped hydro storage	$\overline{PFR}_g^{PS}$	Upper limit of PFR capacity for unit *g*.
PSR	Power shortage rate	*Qspill i,t*	Overflow flow of the reservoir *i*.
PWL	Piecewise Linearization	*qHP i,g, t*	Discharge of hydropower unit *g* at time *t*.
QSFD	Quasi-steady-state frequency deviation	*qPS.G g,t*, *qPS·P g,t*	Generation and pumping flow of the PHS.
RoCoF	rate of change of frequency	*RHP up,t*, *RHP dn,t*	Upward and downward regulation reserve of hydropower unit *g* at time *t*.
SCUC	Security-constrained unit commitment	*RPS.G up,t*, *RPS.G dn,t*	Upward and downward regulation reserve of PHS unit *g* at time *t* in generation mode.
SFR	Secondary frequency regulation	*RPS·P up,t*, *RPS·P dn,t*	Upward and downward regulation reserve of PHS unit *g* at time *t* in pumping mode.
TCUC	Traditional constraint unit commitment	*RHP PFR,t*, *RPS PFR,t*	PFR reserve of hydro and PHS at time *t*.
UHVDC	Ultra-high voltage direct current	*RHP SFR*, *RPS SFR,t*	SFR reserve of hydro and PHS at time *t*.
VRE	Variable renewable energy	*Rreq up,t*, *Rreq dn,t*	Upward and downward reserve demand.
VS-PHS	Variable-speed pumped hydro storage	*Rsys PFR,t*	PFR supply for the system at time *t*.
		$RoCoF_{max}$	Maximum admissible RoCoF.
B. **Indices**		RP_{QSFD}	*QSFD* recovery probability.
t	Index of the time from 1 to N_T.	$SR_{up,t}$	Upward spinning reserve requirement.
k	Index of the scenario from 1 to N_k.	$SR_{dn,t}$	Downward spinning reserve requirement.
i	Index of the hydropower from 1 to N_{HP}.	$T_{R,t}$	Equivalent governor regulation constant.
g	Index of the unit from 1 to $N_{HP,i}/N_{PS}$.	t_{nadir}	Time instance of frequency nadir.
s	Index of the delivery power step.	*uHP i,g,t*	State variable of hydro unit *g* at time *t*.
		uPS.G g,t, *uPS·P g,t*	Binary variable indicating PHS unit state.
C. **Parameters and variables**		*VHP i,t*	Reservoir capacity of hydropower station *i*.
a_v, b_v, c_v, d_v	Coefficients of the PWL constraints.	*Vup t*	Upstream reservoir capacity of the PHS.
BS_{WT},BS_{PV}	Wind and PV output interpolation function.	*Vup min*, *Vup max*	Minimum and maximum reservoir capacity.
Cap_g	Installed capacity of unit *g*.	v_{in}, v_{out}, v_r	Cut-in, cut-out, and rated wind speed.
c_{cur}, c_{loss}, c_{spill}	Penalty costs for curtailment, spillage, and power shortage, respectively.	VD_{RoCoF}	*RoCoF* violation gap.
cHP peak, *cPS peak*	Peak shaving reserve cost.	$VD_{\Delta fnadir}$	*MFD* violation gap.
cHP PFR, *cPS PFR*	PFR reserve cost.	$z_{\alpha/2}$	Quantile of the standard normal distribution.
cHP SFR, *cPS SFR*	SFR reserve cost.	*ZHP i,t*, *Ztail i,t*	Fore-bay and tailrace water level.
CI_{WT}, CI_{PV}	Power output confidence intervals.	*Δfmax nadir*, *Δfmax ss*	Maximum admissible frequency deviation at the naidr and quasi-steady state.
cHP on, *cPS on*	Start-up cost of hydro and PHS units.	*ΔPPS ramp, g*	Ramping speed of the PHS unit *g*.
e_p	Adjustment coefficient.	*ΔPHP ramp,i,g*	Ramping speed of unit *g* at hydro station *i*.
f_0	Nominal frequency of the system.	Δt	Time step.
FPRC k,t, *FFRC k,t*	Reserve opportunity cost of peak shaving and frequency regulation.	$\xi_{HP,i}$	Hydraulic conversion coefficients of the hydropower station *i*.
FRisk k,t, *Fon k,t*	Total risk and the unit start-stop cost.	*ξPS G*, *ξPS P*	Hydraulic conversion coefficient of PHS.
F_H	System equivalent turbine parameter.	$\overline{v}$	Maximum number of segments for PWL.
fZV i, *fZQ i*	Reservoir water level-storage capacity and tailwater level-discharge curve.	ω_n	Natural frequency.
$H_{sys,t}$	System inertia at time *t*.	ξ	Damping ratio.
Hmin sys,t	Minimum inertia requirement at time *t*.	σ_{WT}, σ_{PV}	Standard deviation of wind and PV output.
$H_{i,g,t}$	Hydraulic head of unit *g* in hydro station *i*.	ε_{DC}	Precision control index of CHIDS's output.
H_{ave}	Average hydraulic head of the PHS.	γ, δ	PWL auxiliary variable.
k_d	Load damping rate.	*μmax VRE*	Upper limit of the VRE curtailment rate.
$k_{f,g}^{PS}$, *kHP f,i,g*	PFR capability coefficient.	η_{VRE}	VRE penetration level.

Data availability

Data will be made available on request.

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